# Convex and concave decomposition of digital shapes using plane probing and visibility

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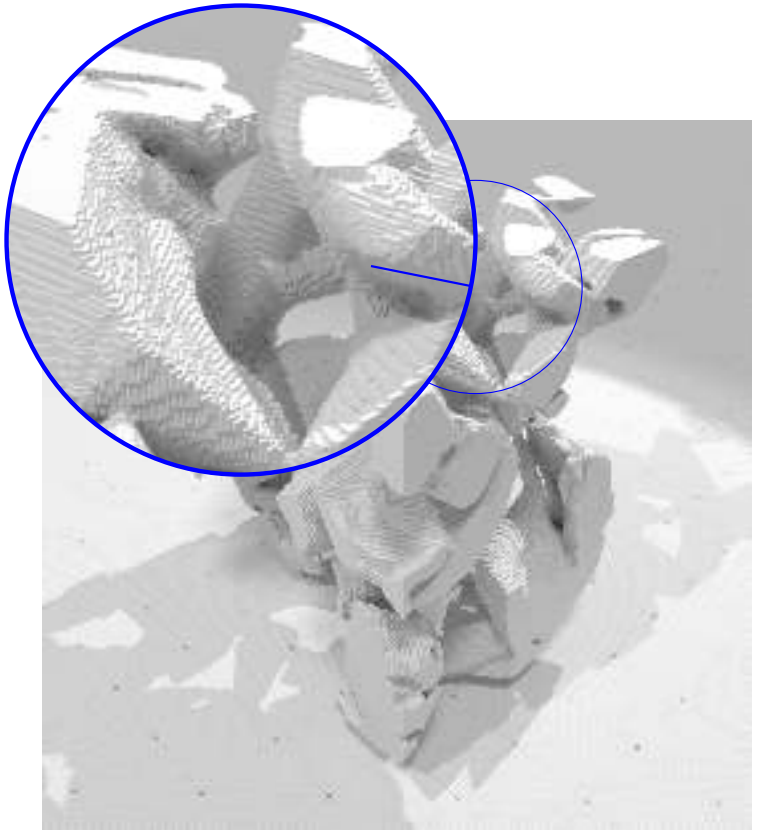
Discrete Geometry and Mathematical Morphology (DGMM 2025)
University of Groningen, the Netherlands

# Context: geometry of shapes in 3D imaging

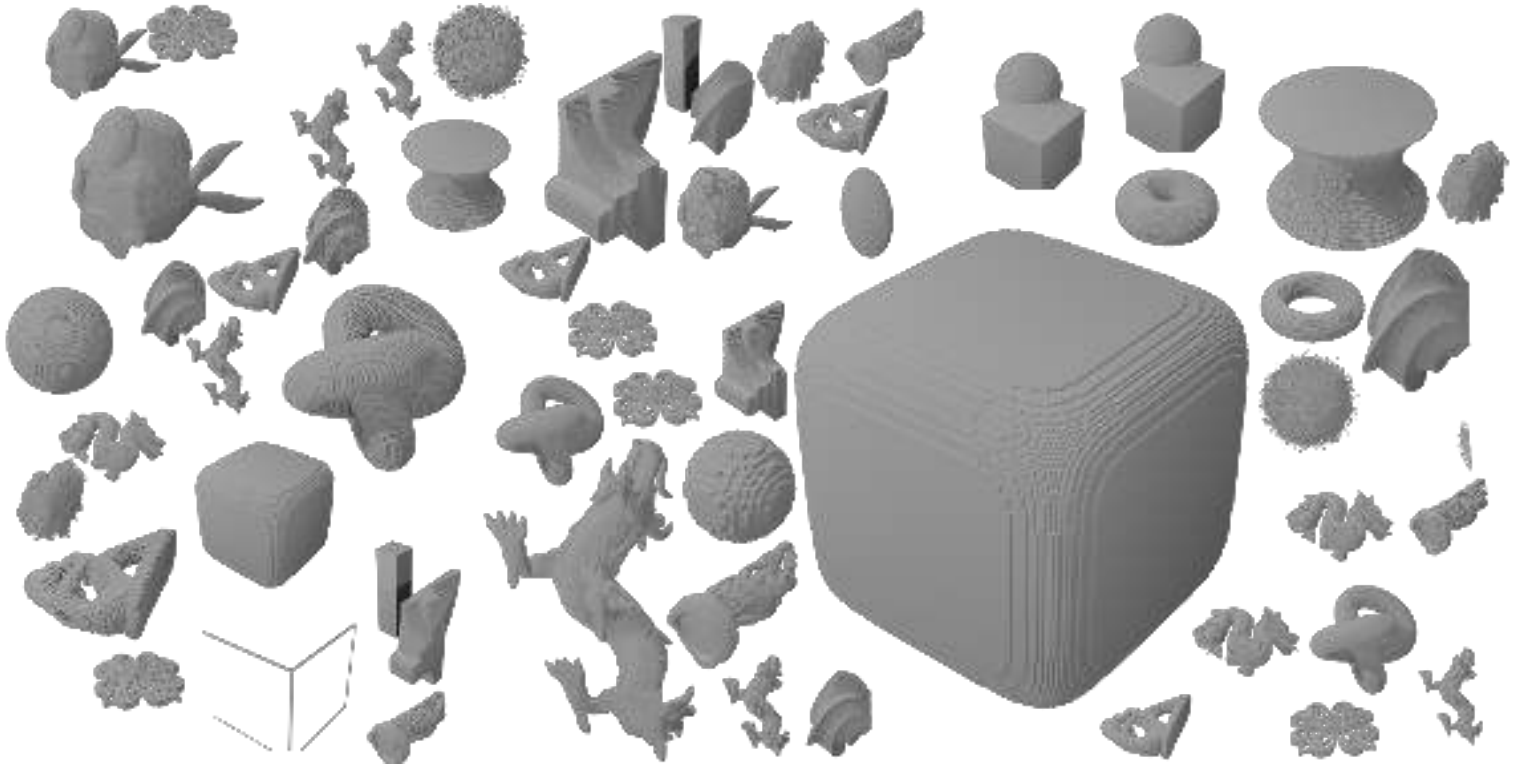
MRI, CT-scan, PET-scan, confocal microscopy, ...



snow micro-tomography

# Context: geometry of shapes in 3D imaging

geometric modeling, shape indexing, machine learning, ...

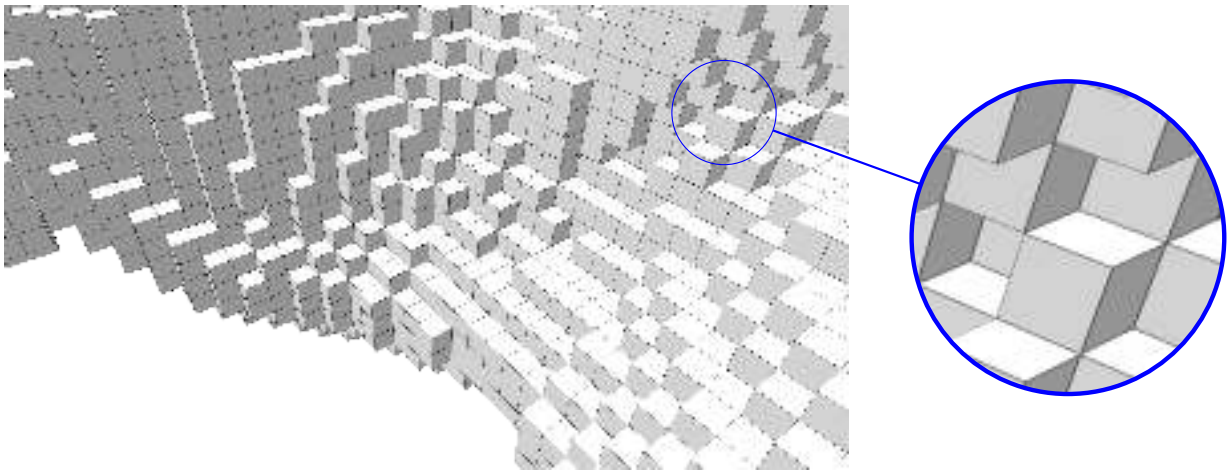


(https://github.com/dcoeurjo/VolGallery)

# Digital shapes and surfaces

**Definitions**

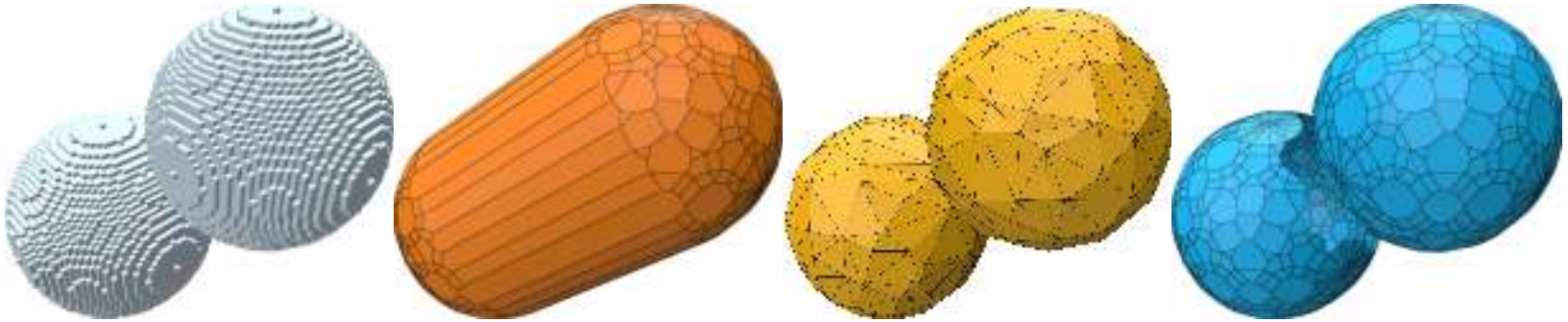
Digital shape $Z$ = set of connected *voxels*
Digital surface $\partial Z$ = boundary of $Z$, set of *surfels*



**Properties**

| | |
|---|---|
| topology | closed, oriented, but non manifold in general |
| geometry | subset of a cubical grid, integer points |

# How to identify/represent convex and concave parts?



2 balls $B_{25} \cap \mathbb{Z}^3$ the convex hull an approximation local convexity

- identify vertices that are locally extremal in some direction
- identify edges and faces joining them
  - edges should form convex angles, faces around vertices should form convex cones
  - edges and faces should stay close to the digital surface, without crossing it

# Outline

Definition of local convexity/concavity based on visibility

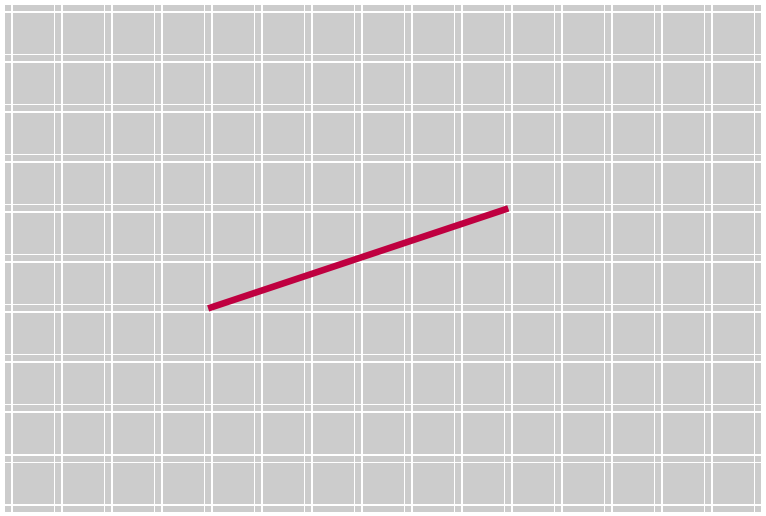
Algorithm to reconstruct convex/concave parts

# Framework

## Cubical grid

- cubical grid $\mathcal{C}^d$: partition of $\mathbb{R}^d$, where every cell is a cartesian product of $d$ intervals of the form $\{x\}$ (closed) or $(x, x+1)$ (open)
- the dimension of a cell is the number of open intervals
- $\mathcal{C}_k^d$ denotes the set of $k$-dimensional cells ($k$-cells), $\mathcal{C}_0^d = \mathbb{Z}^d$

# Basic topological notions



**Cover and star of $Y \subset \mathbb{R}^d$**

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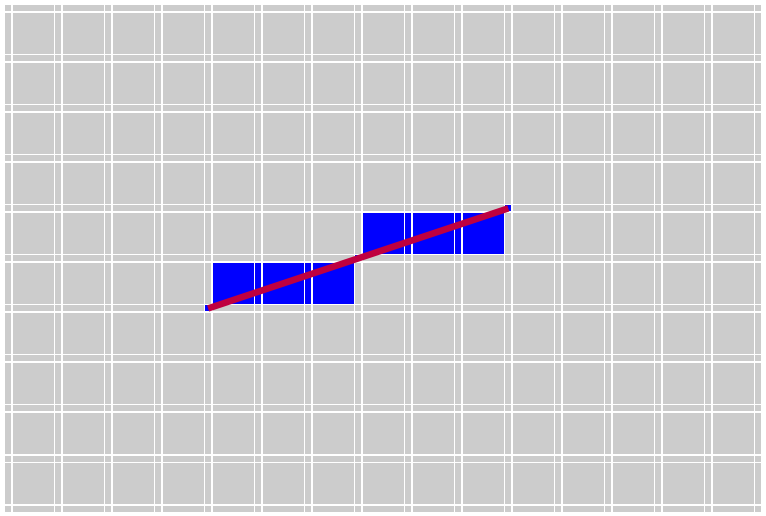

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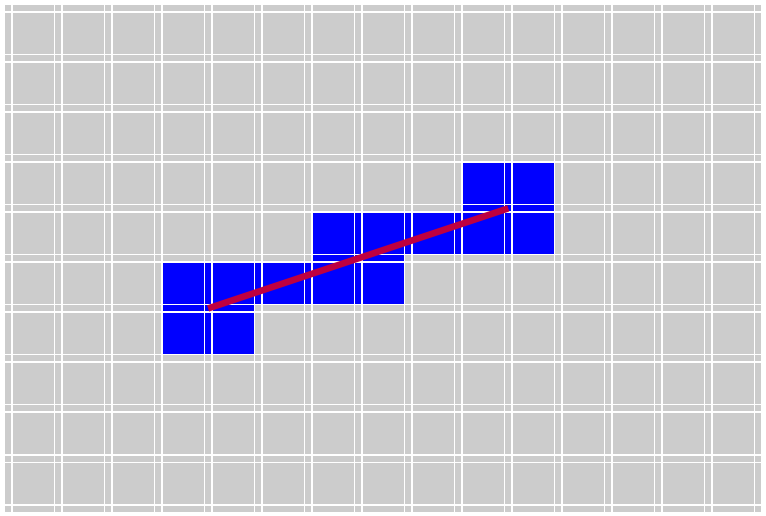

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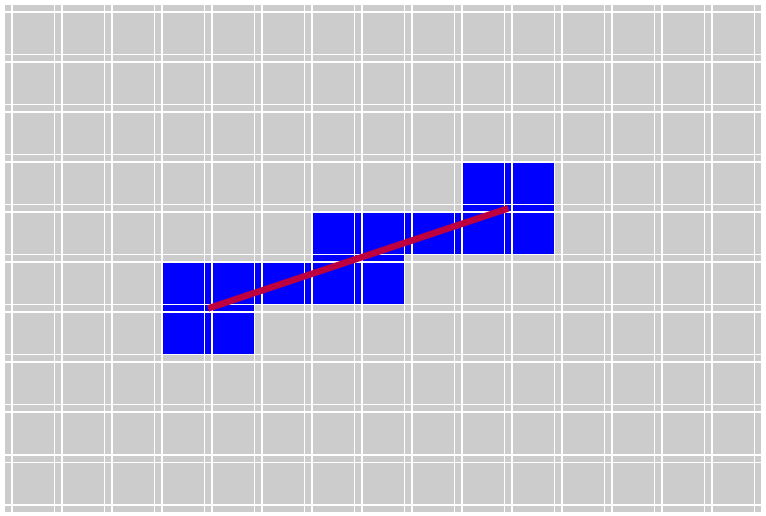

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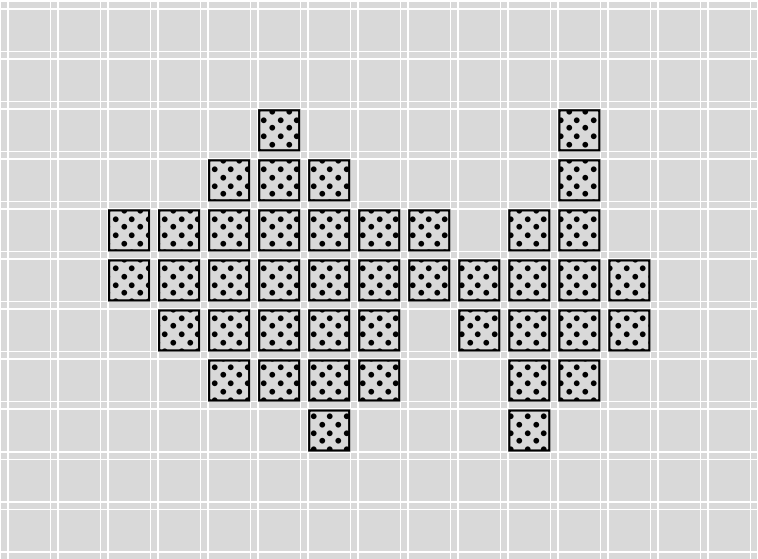
# Basic definitions



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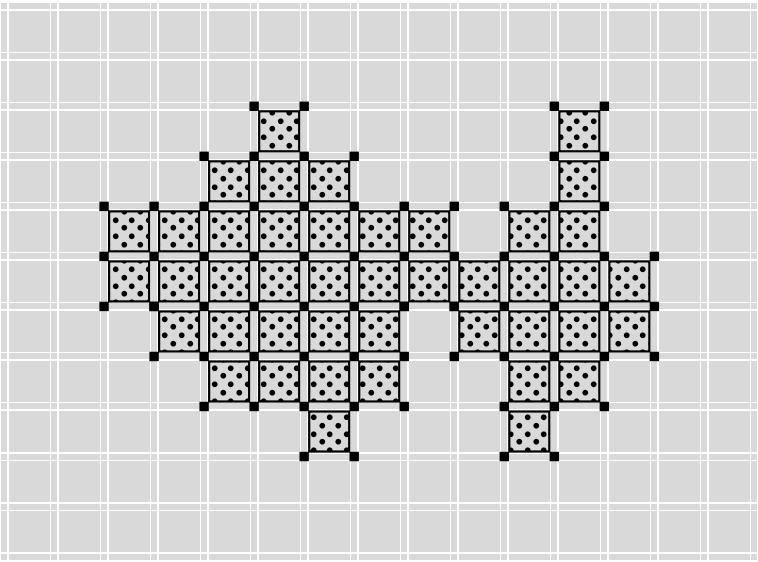

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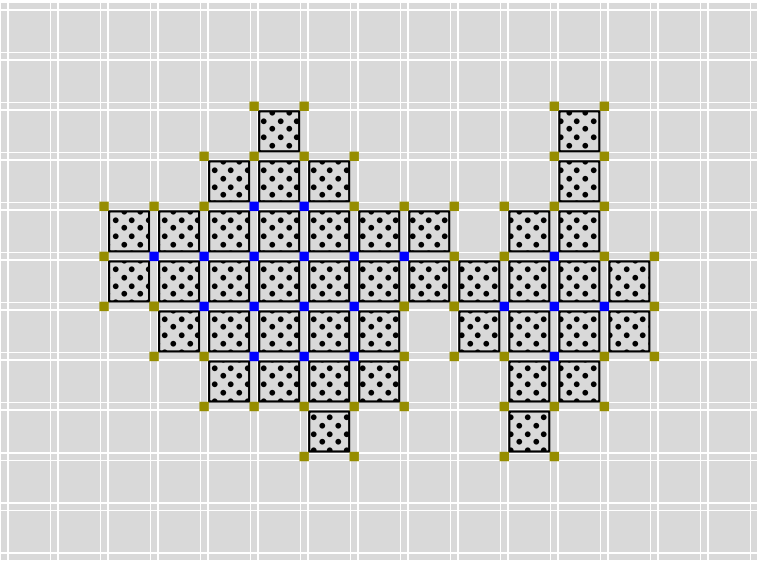

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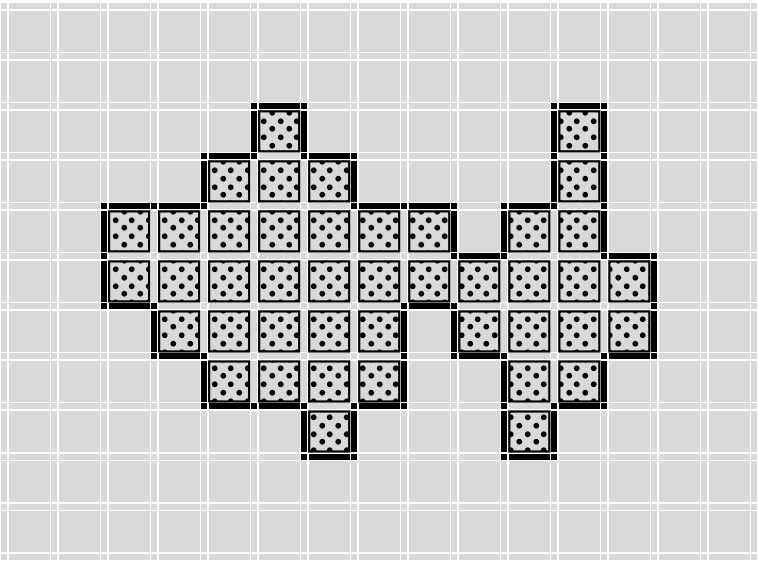

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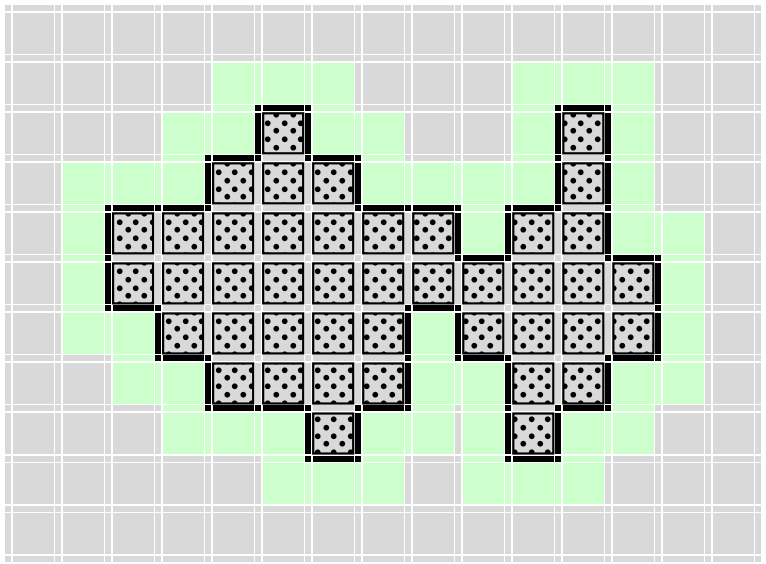

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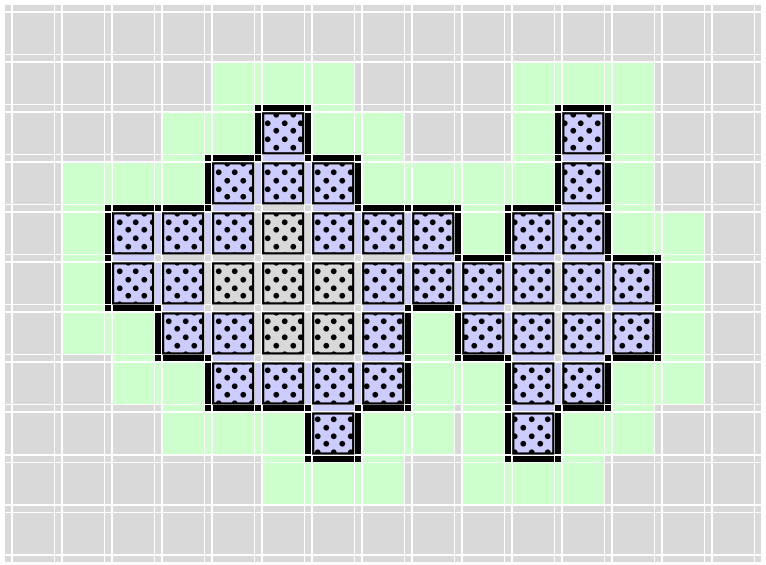

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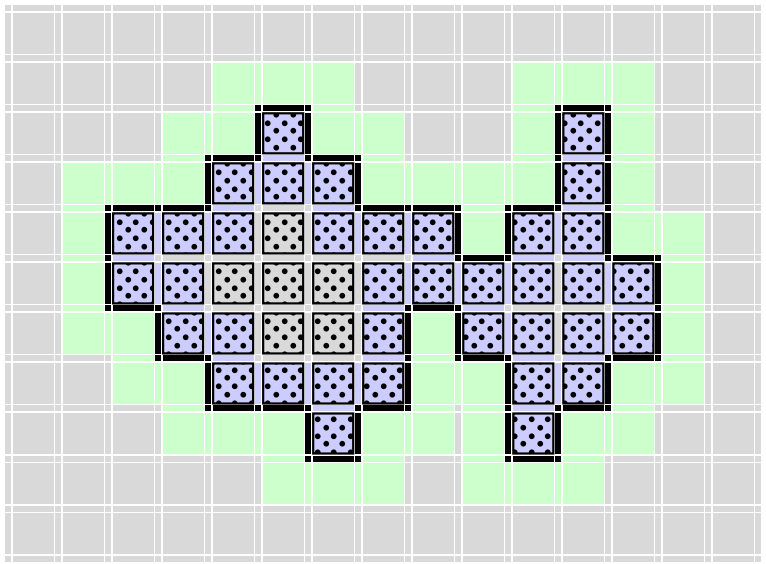

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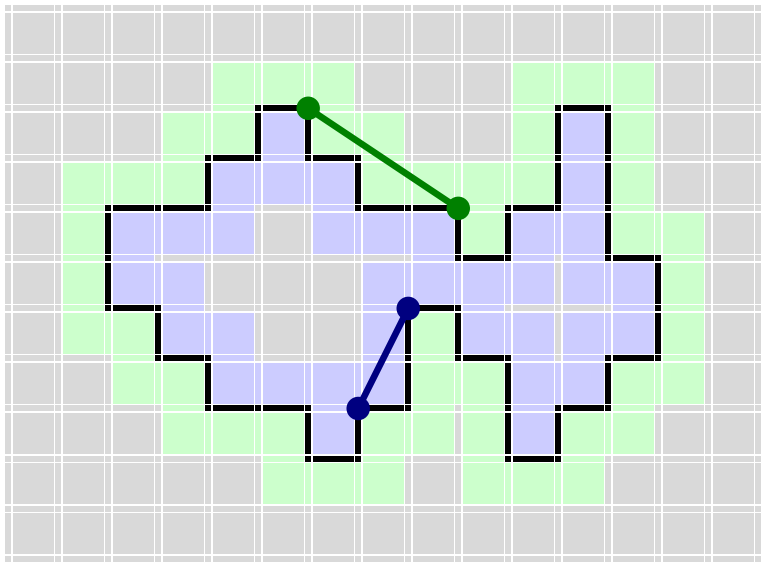
We have $\mathrm{Star}\,(\mathrm{Bd}\,(K)) = \mathrm{Bd}\,(K) \sqcup \mathrm{Out}\,(K) \sqcup \mathrm{In}\,(K)$

# Convex and concave visibility



**Convex $K$-visibility**

$A := \{p_1, \ldots, p_n\} \subset Z_K$ is **convex $K$-visible** iff $\mathrm{Cover}\,(\mathrm{Cvxh}\,(A)) \subset \mathrm{Out}\,(K) \cup \mathrm{Bd}\,(K)$

**Concave $K$-visibility**

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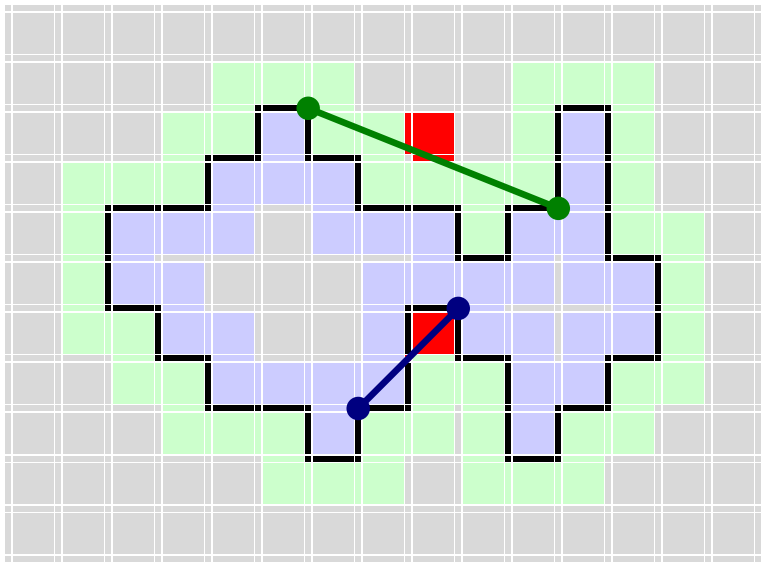

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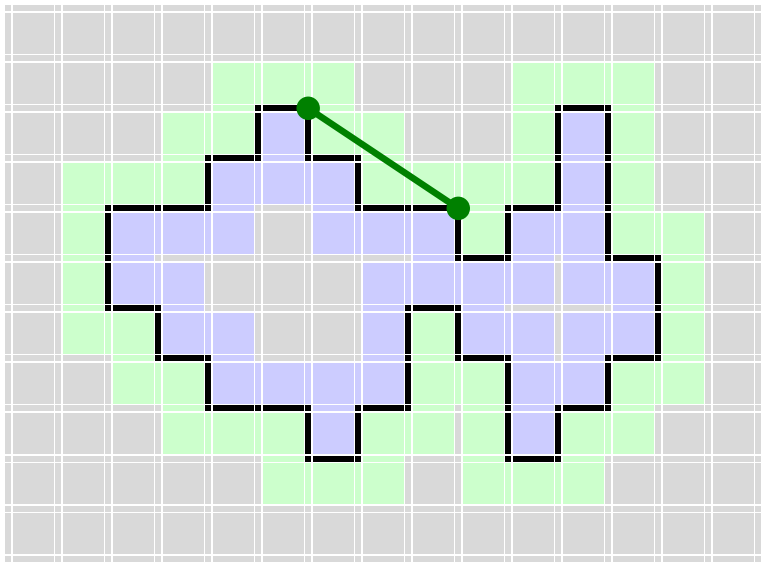

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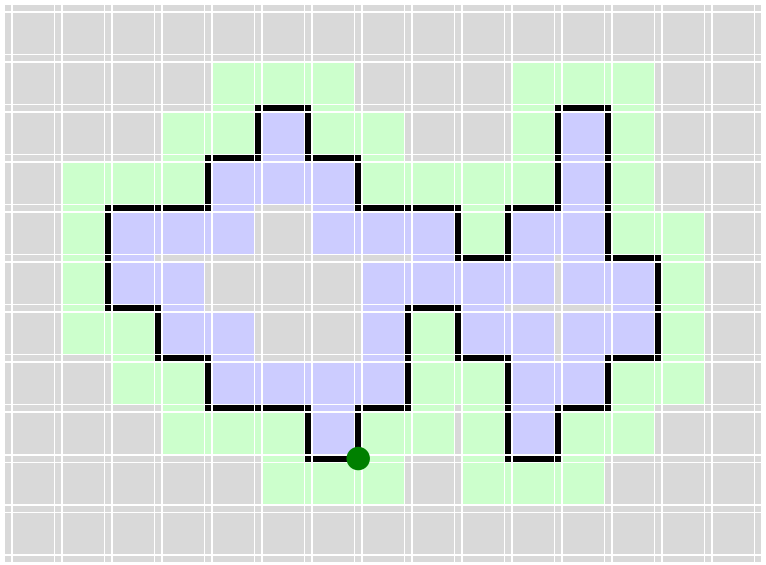
From now on, focus on convex visibility (concave visibility is entirely symmetric).

# $K$-visibility cone, locally-convex point, edge, face, ...



**$C_K(p)$**

The **convex $K$-visibility cone** $C_K(p)$ of $p$ is the set of points $q \in Z_K$ with $\{p, q\}$ convex $K$-visible.

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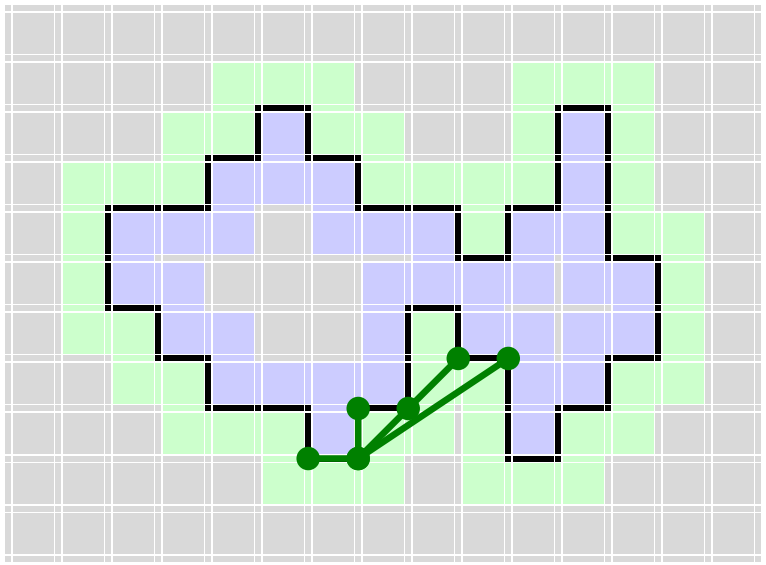

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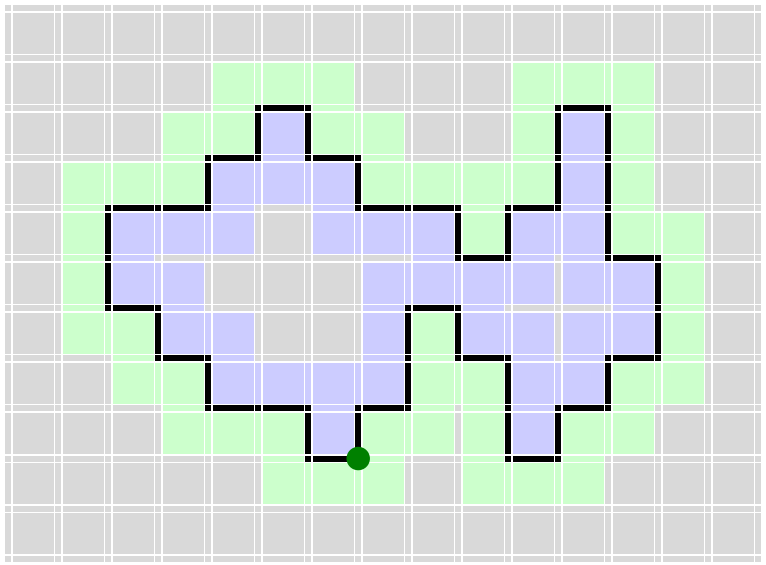

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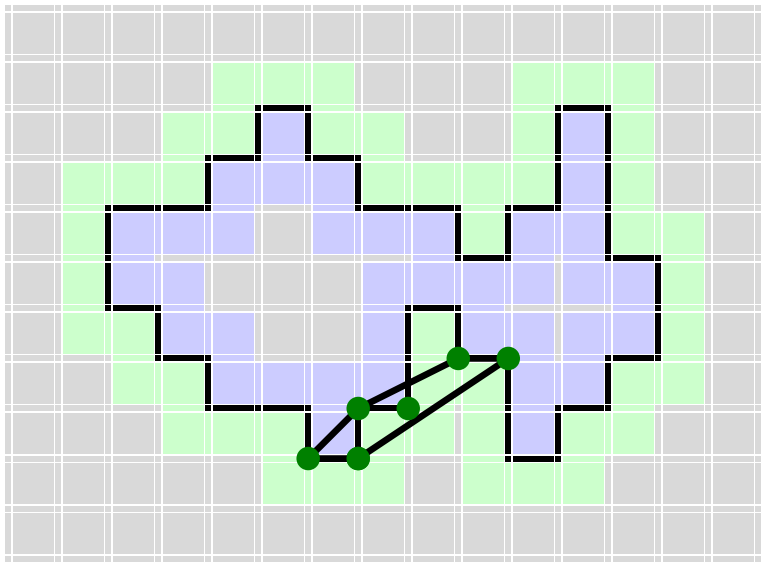

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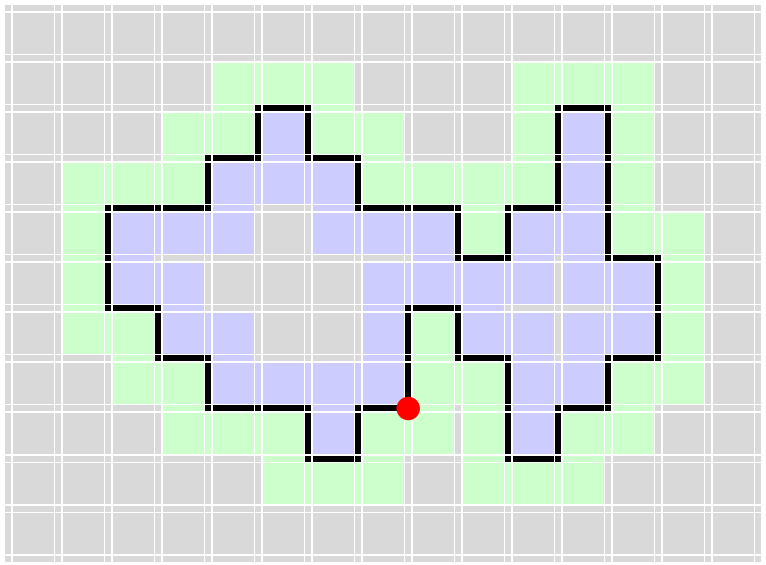

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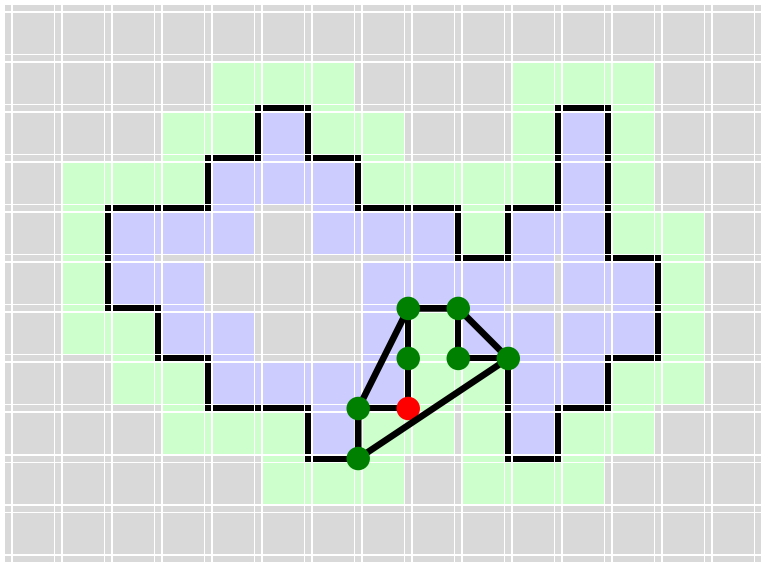

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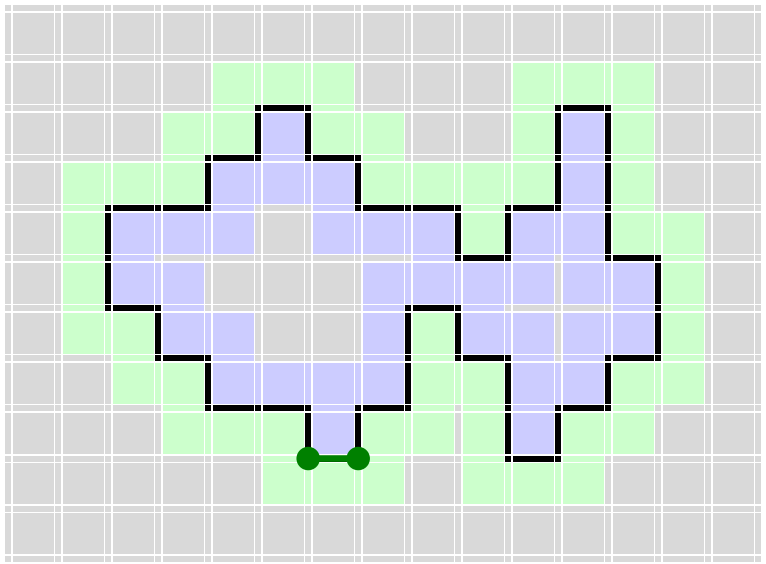

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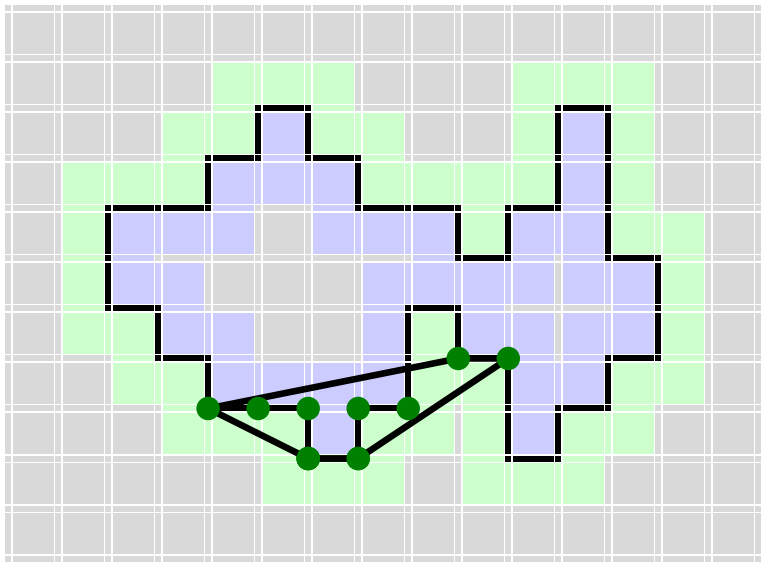

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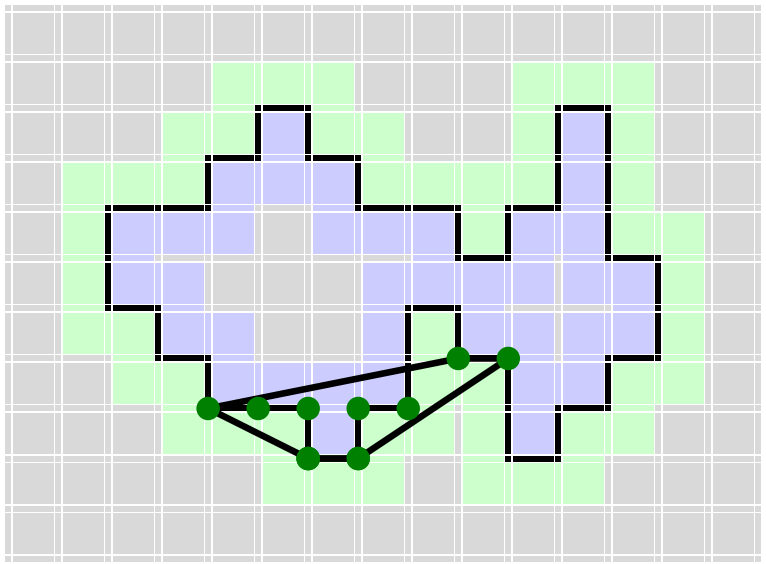

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## Lemma (Consistency of local convexity)

*If $F$ is locally convex in $K$, then any subset of $F$ is locally convex in $K$.*

# Full convexity implies local convexity

**Full convexity** [Lachaud. 2021]

The digital set $X \subset \mathbb{Z}^d$ is fully convex iff $\mathrm{Star}\,(\mathrm{Cvxh}\,(X)) \subset \mathrm{Star}\,(X)$.

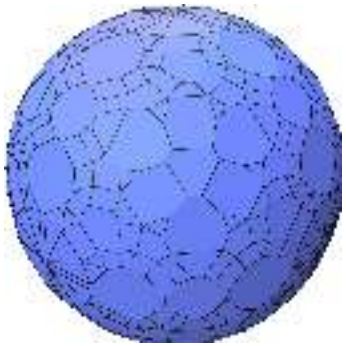
- full convexity implies classical digital convexity
- full convexity implies connectedness

**Theorem**

*Let $K \subset \mathcal{C}_d^d$ and $X_K$ fully convex. The vertices and the faces of $\mathrm{Cvxh}\,(X_K)$ are locally convex vertices and locally convex faces of $K$.*

$X_K$

$\mathrm{Cvxh}\,(X_K)$



locally convex faces

# Outline

Definition of local convexity/concavity based on visibility

Algorithm to reconstruct convex/concave parts

# Algorithm to reconstruct convex parts

**Input** a set of $d$-dimensional cells $K$

- compute the boundary 0-cells $Z_K$ of $K$
- compute the visibility cones $C_K(p)$, for all $p \in Z_K$
- for all point $p \in Z_K$
  - check if $p$ is locally convex by computing $\mathrm{Cvxh}\,(C_K(p))$
  - collect incident edges in $E$ if it is the case
  - store $p$ in $V$ if it is the case
- for all edge $e := (p_1, p_2) \in E$
  - check if $e$ is locally convex by computing $\mathrm{Cvxh}\,(\cup_i C_K(p_i))$
  - collect incident faces in $F$ if it is the case
- for all face $f := (p_1, \ldots, p_k) \in F$
  - check if $f$ is locally convex by computing $\mathrm{Cvxh}\,(\cup_i C_K(p_i))$
  - and store it in $G$ if it is the case
- return locally convex points $V$ and faces $G$

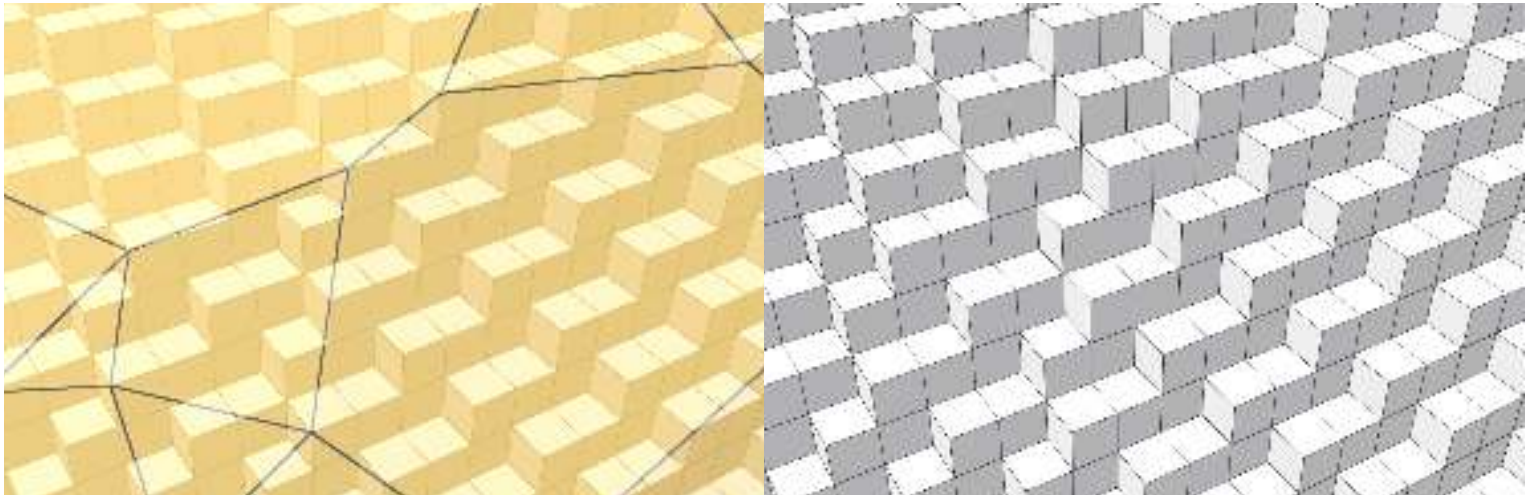
# Algorithm to reconstruct convex parts

**Input** a set of $d$-dimensional cells $K$

- compute the boundary 0-cells $Z_K$ of $K$
- compute the visibility cones $C_K(p)$, for all $p \in Z_K$
- for all point $p \in Z_K$
  - check if $p$ is locally convex by computing $\mathrm{Cvxh}\,(C_K(p))$
  - collect incident edges in $E$ if it is the case
  - store $p$ in $V$ if it is the case
- for all edge $e := (p_1, p_2) \in E$
  - check if $e$ is locally convex by computing $\mathrm{Cvxh}\,(\cup_i C_K(p_i))$
  - collect incident faces in $F$ if it is the case
- for all face $f := (p_1, \ldots, p_k) \in F$
  - check if $f$ is locally convex by computing $\mathrm{Cvxh}\,(\cup_i C_K(p_i))$
  - and store it in $G$ if it is the case
- return locally convex points $V$ and faces $G$

More than 95% of the time is spent in computing visibility cones.
$\Rightarrow$ We have to prune the input set $Z_K$.

# Discarding non-extremal points



**Objective**

Quickly discard points of $Z_K$ that cannot be locally convex points, without computing their visibility cone.

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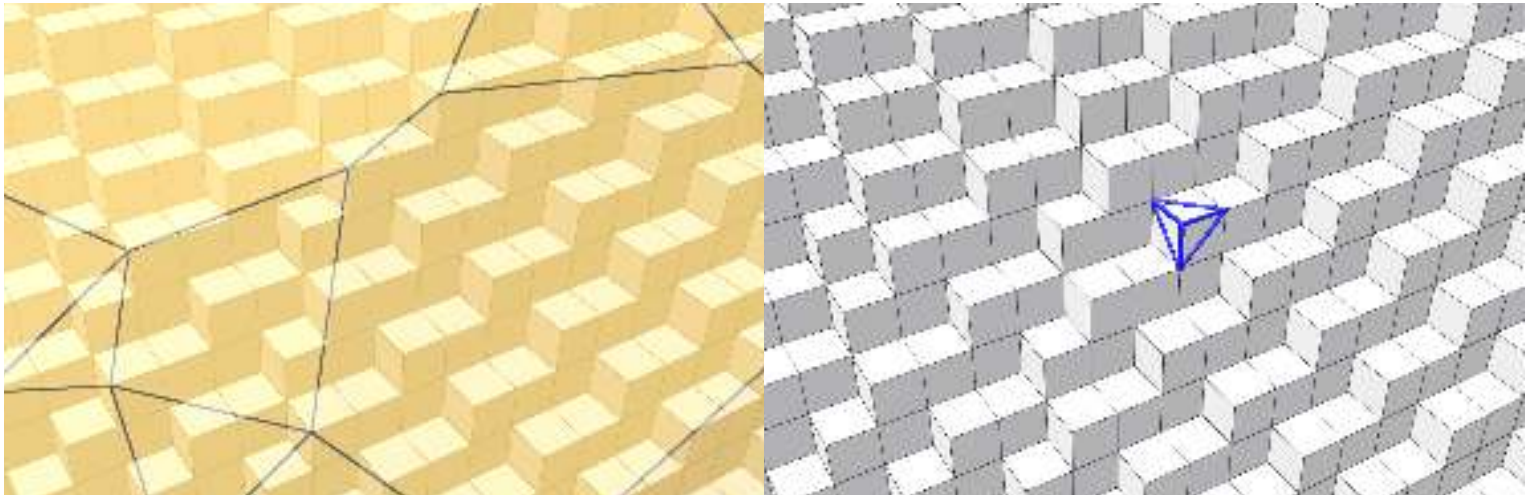

**Objective**

Quickly discard points of $Z_K$ that cannot be locally convex points, without computing their visibility cone.

- keep only the salient corners of $Z_K$

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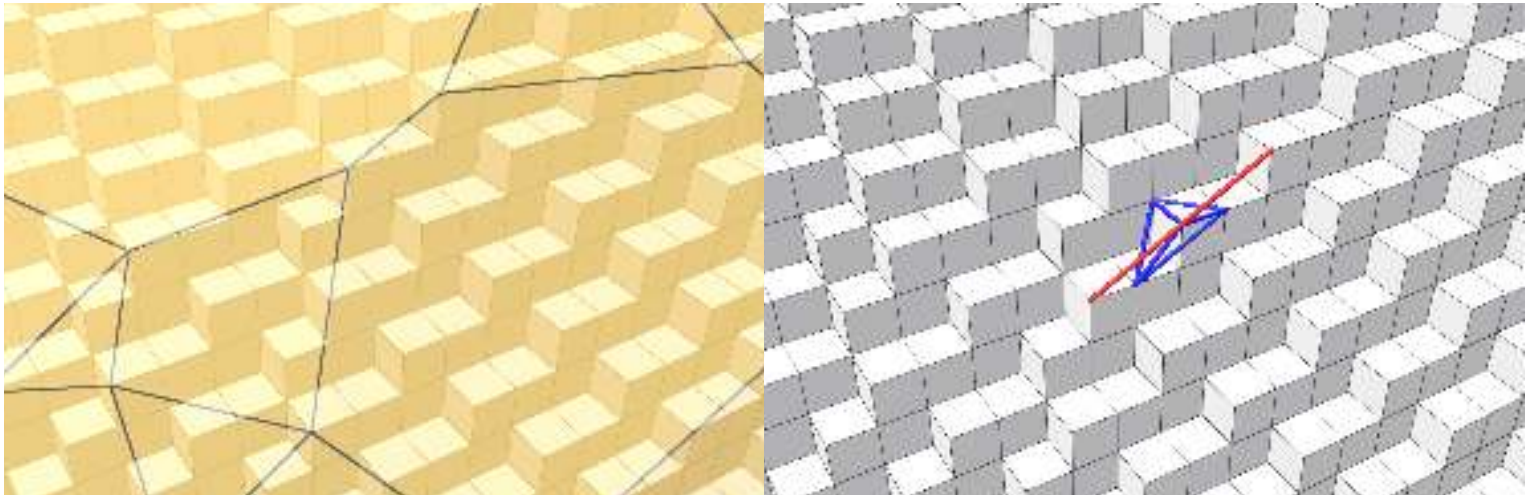

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Quickly discard points of $Z_K$ that cannot be locally convex points, without computing their visibility cone.

- keep only the salient corners of $Z_K$
- eliminate corners $c$ that are in-between two points of $Z_K$, i.e., $\exists \mathbf{v} \in \mathbb{Z}^d$ with $c \pm \mathbf{v} \in Z_K$

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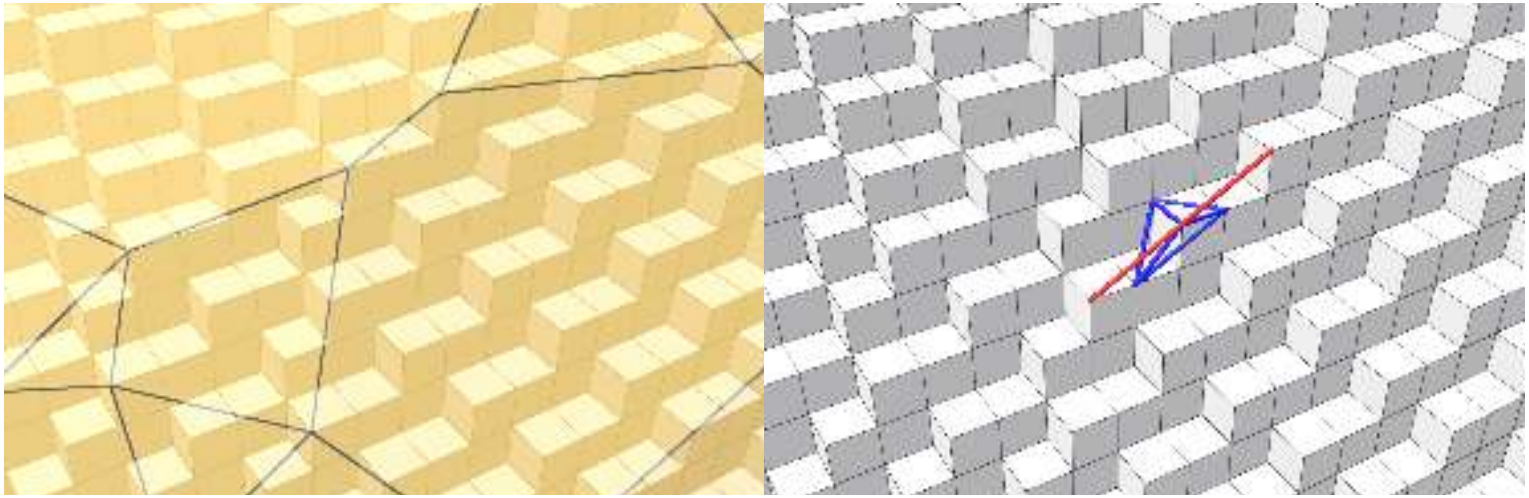

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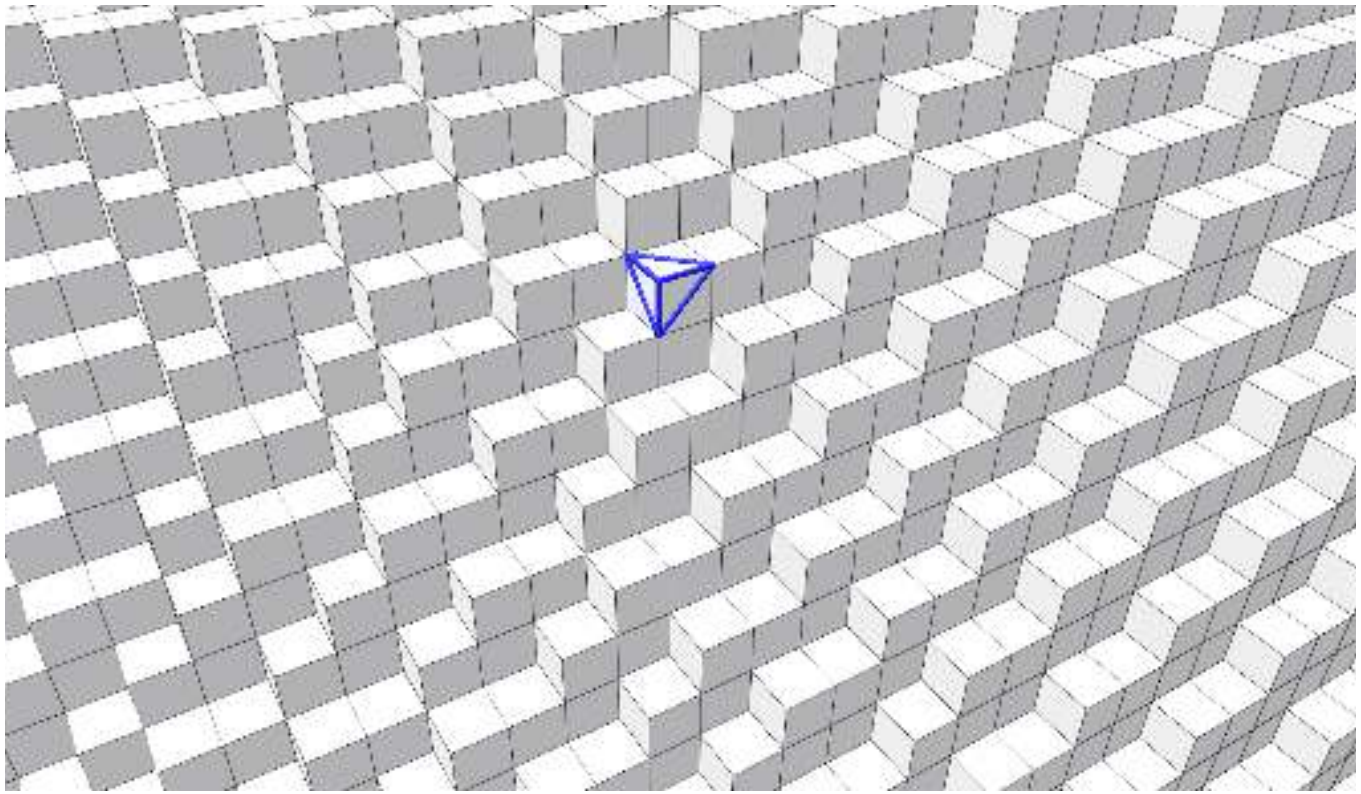
- method to find $\mathbf{v}$: a variant of plane probing, normally used for plane recognition [Lachaud, Provençal, Roussillon. 17]

# Illustration of our algorithm IsExtremal



- Tetrahedron with one vertex fixed (the point to test), described by a matrix $\mathbf{M} = [\mathbf{m}_1, \mathbf{m}_2, \mathbf{m}_3]$
- Initialize $\mathbf{M}$ with $[\pm\mathbf{e}_1, \pm\mathbf{e}_2, \pm\mathbf{e}_3]$ and update it

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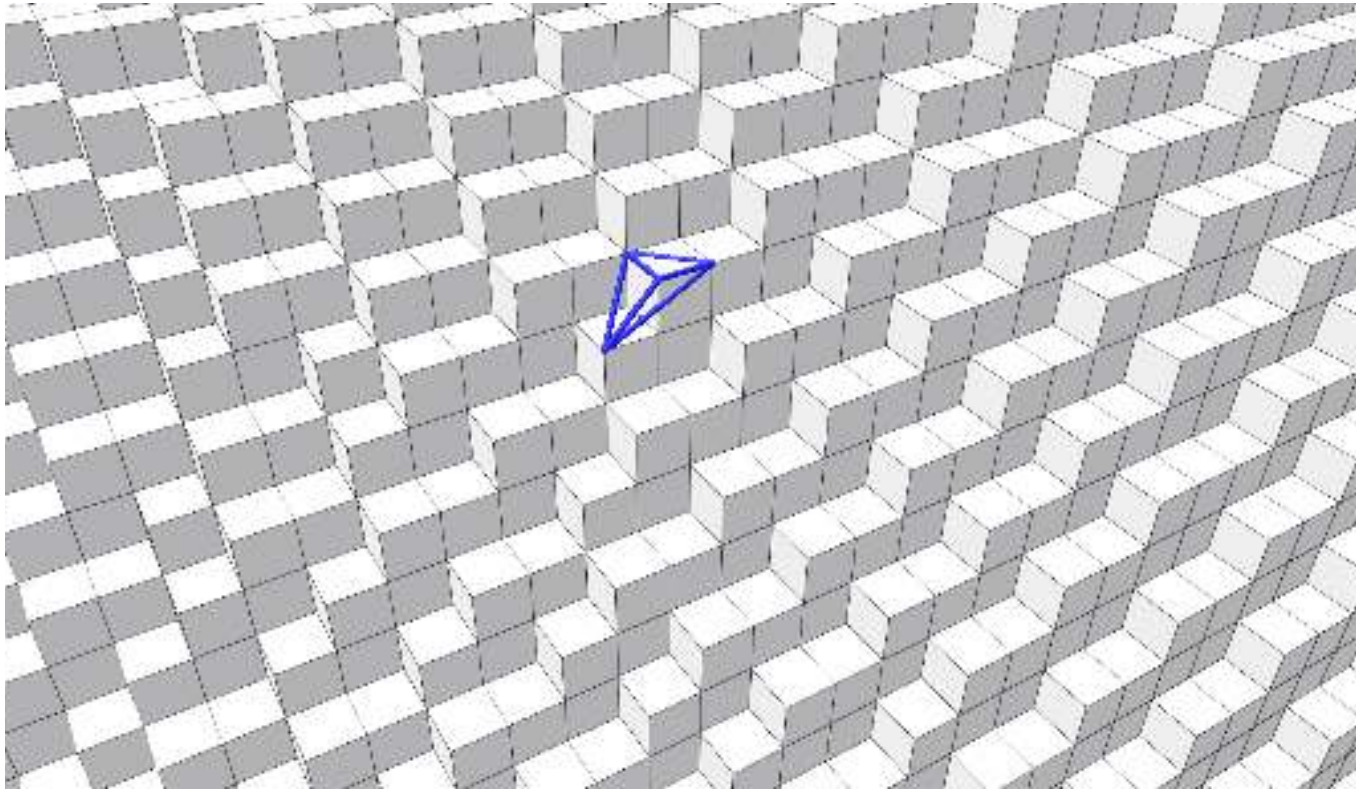

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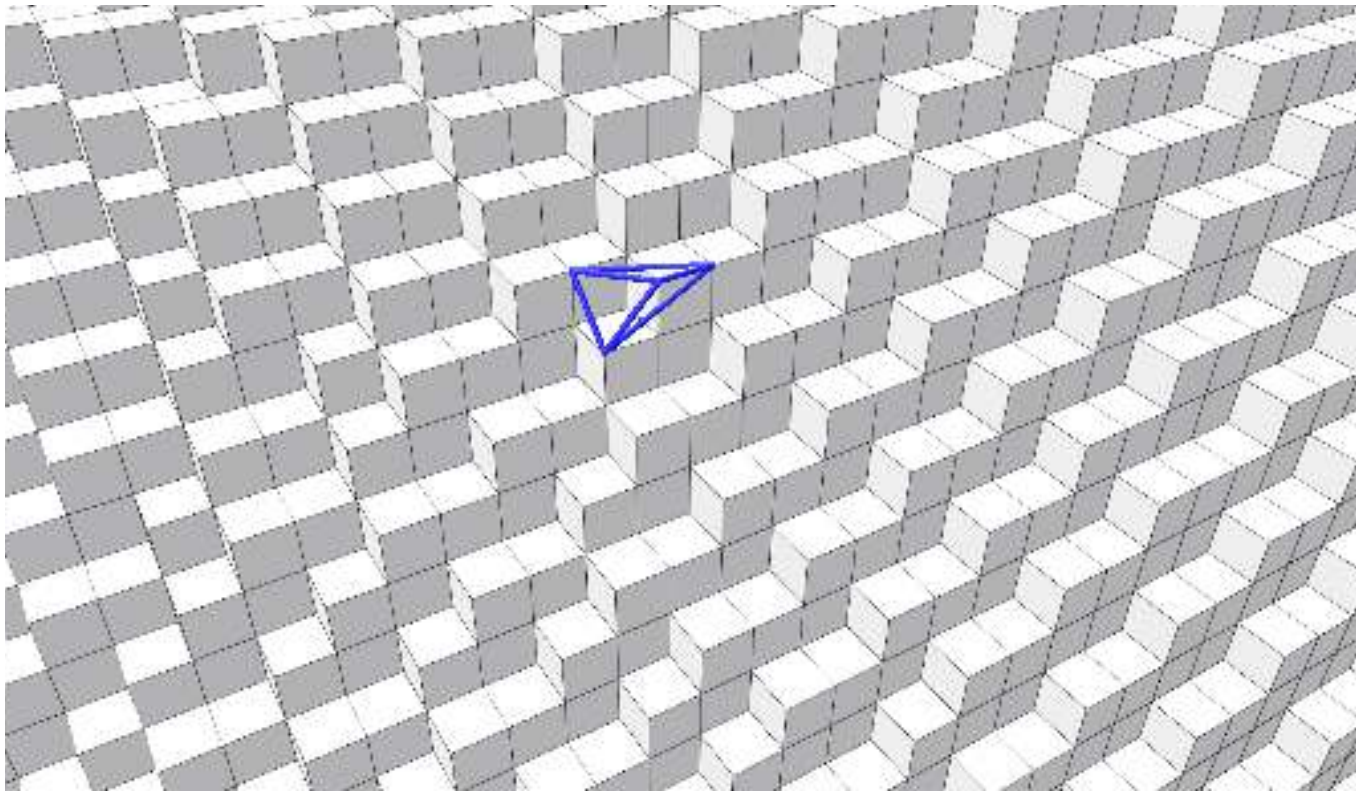

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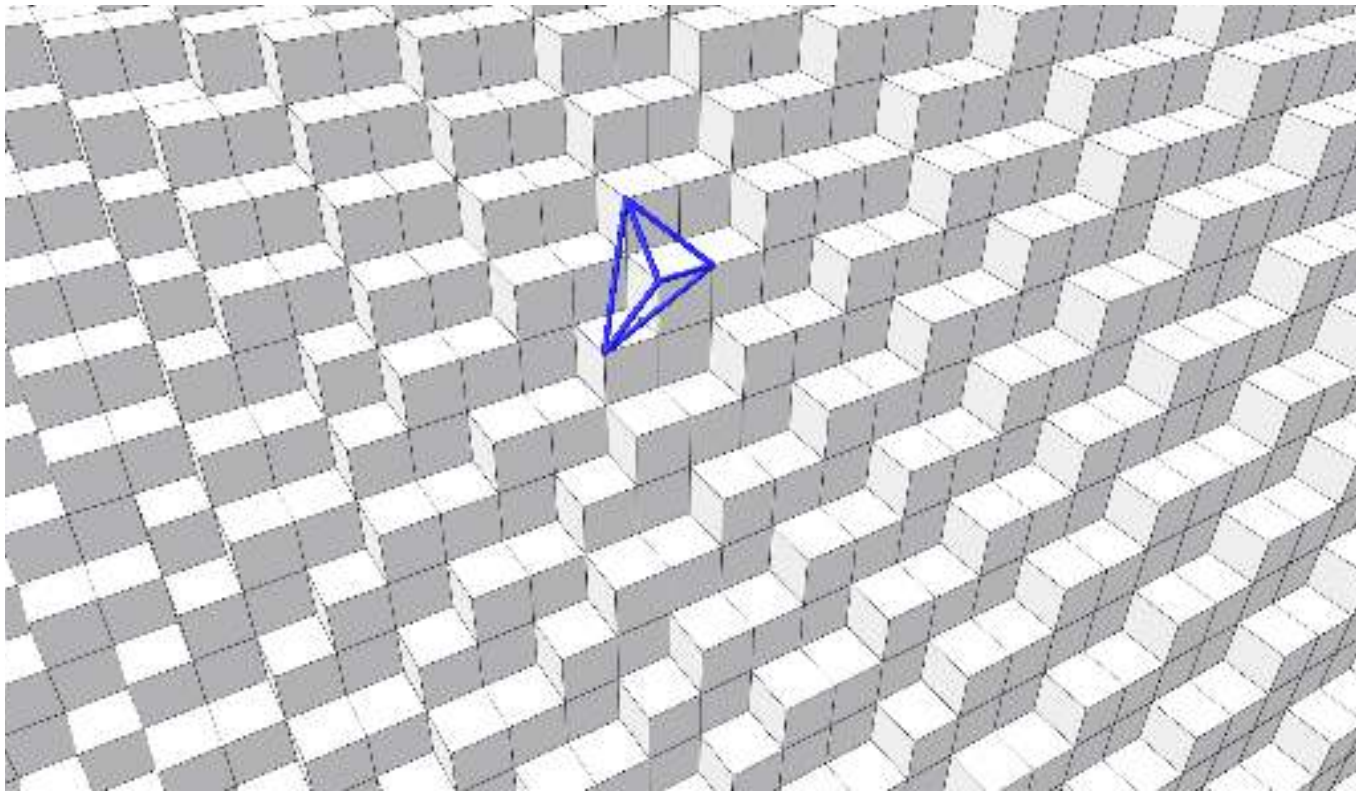

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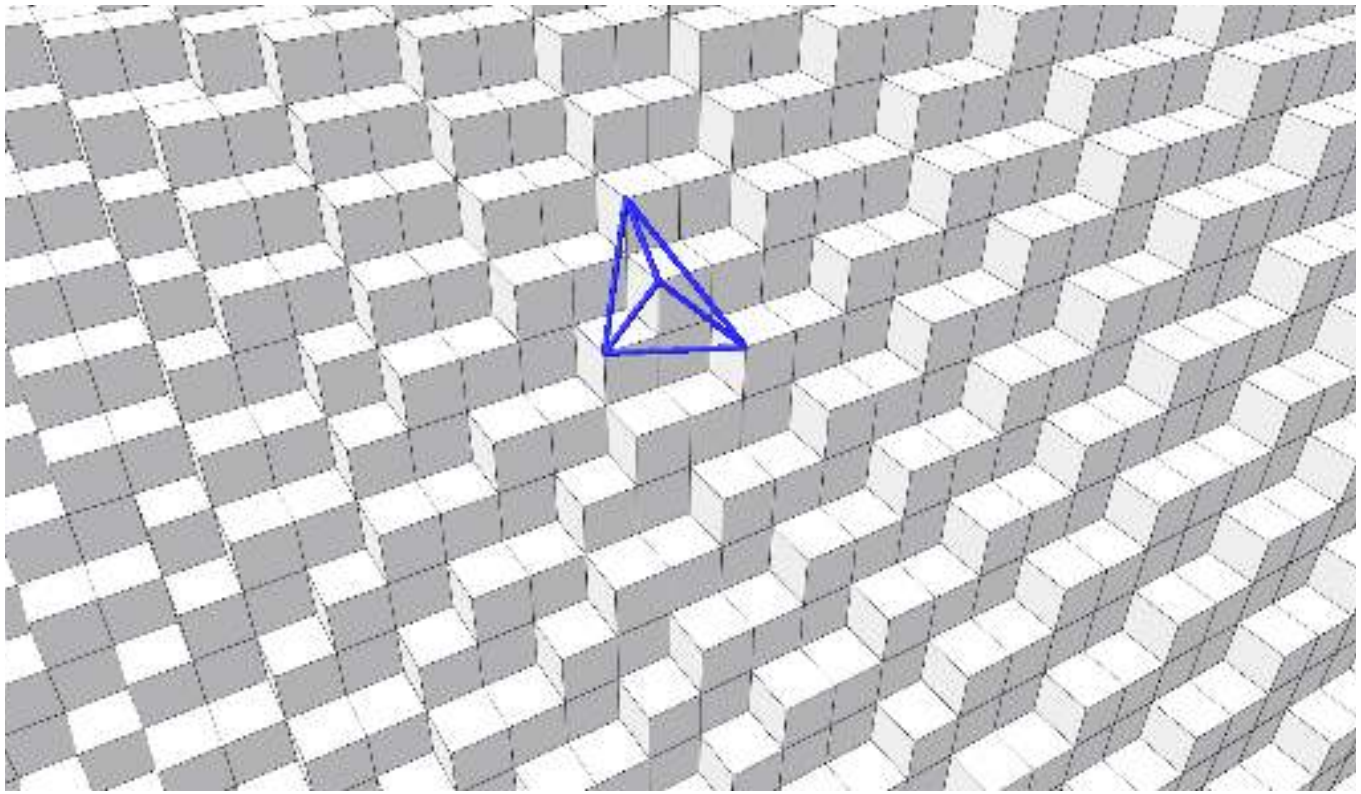

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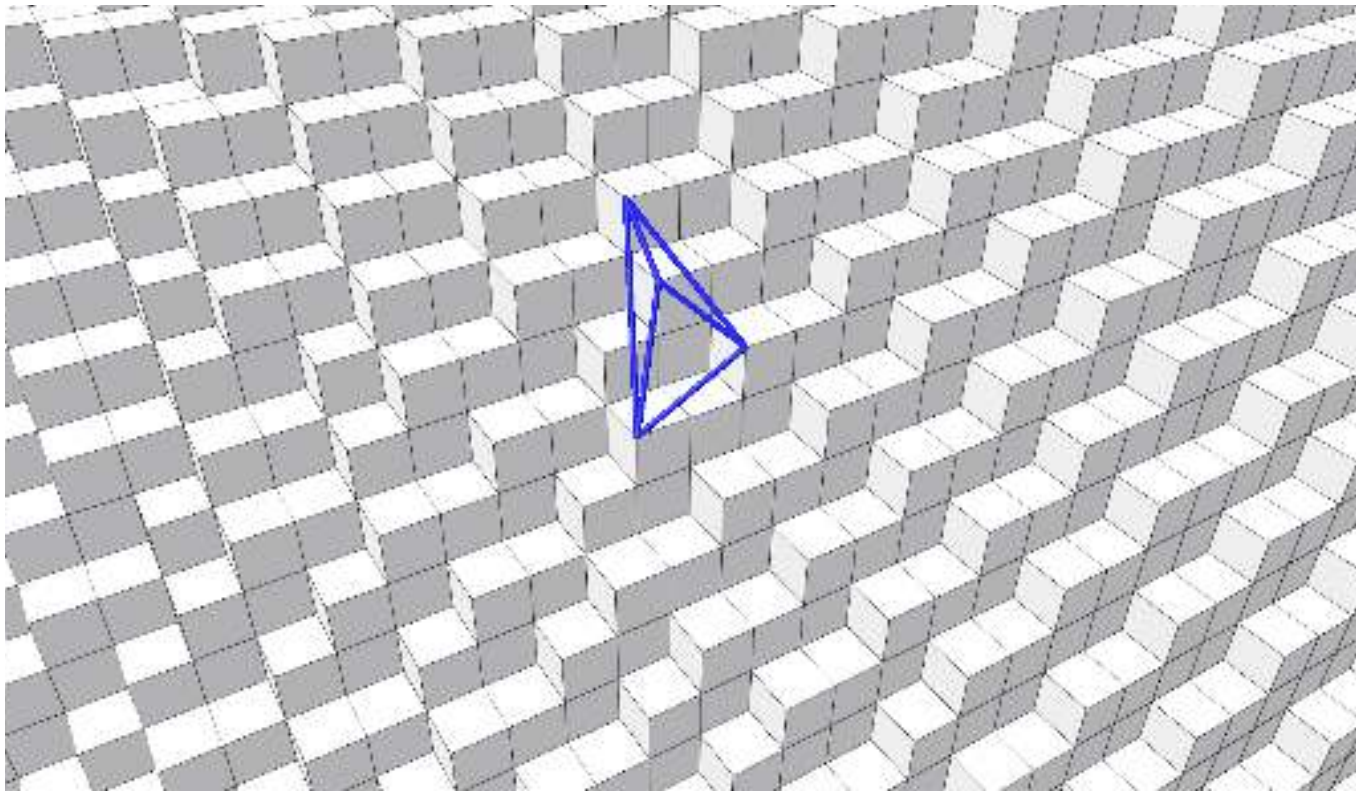

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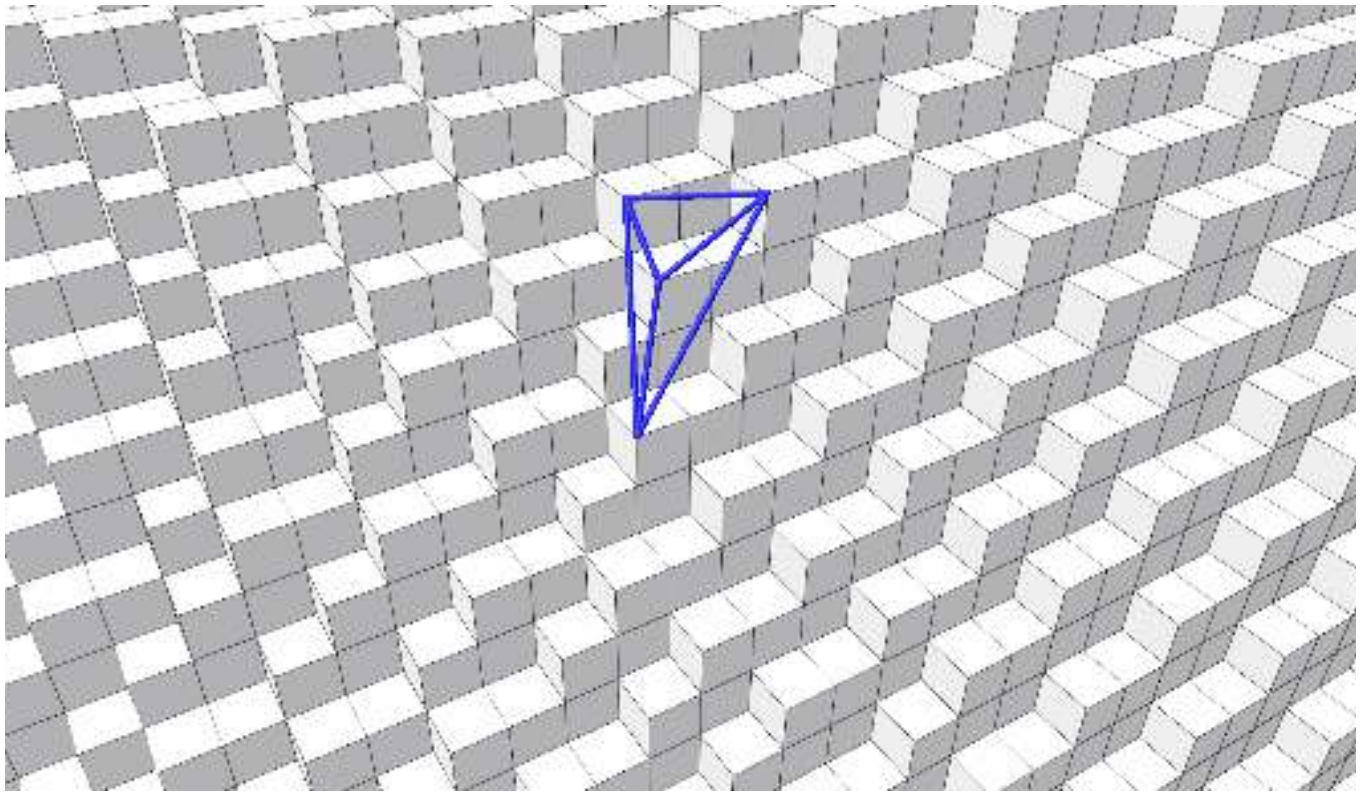

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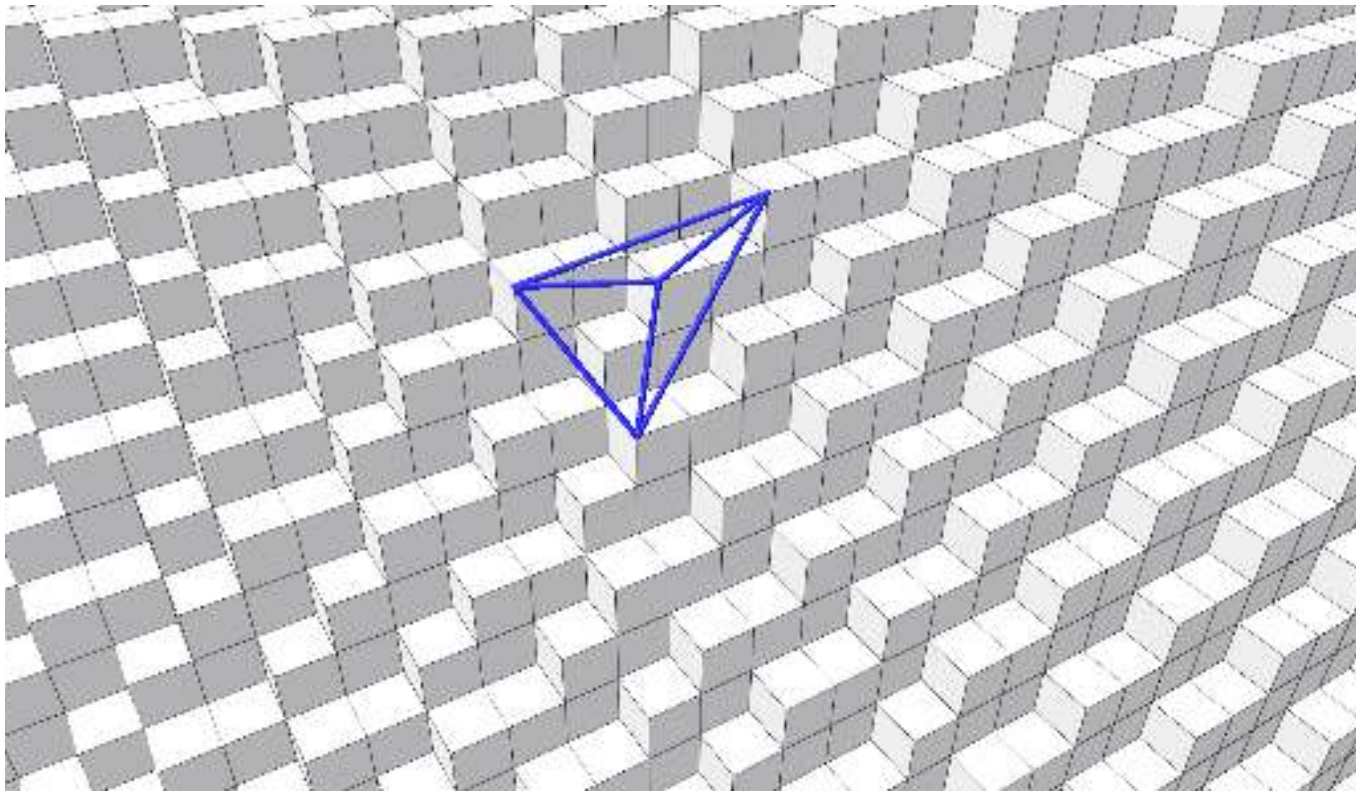

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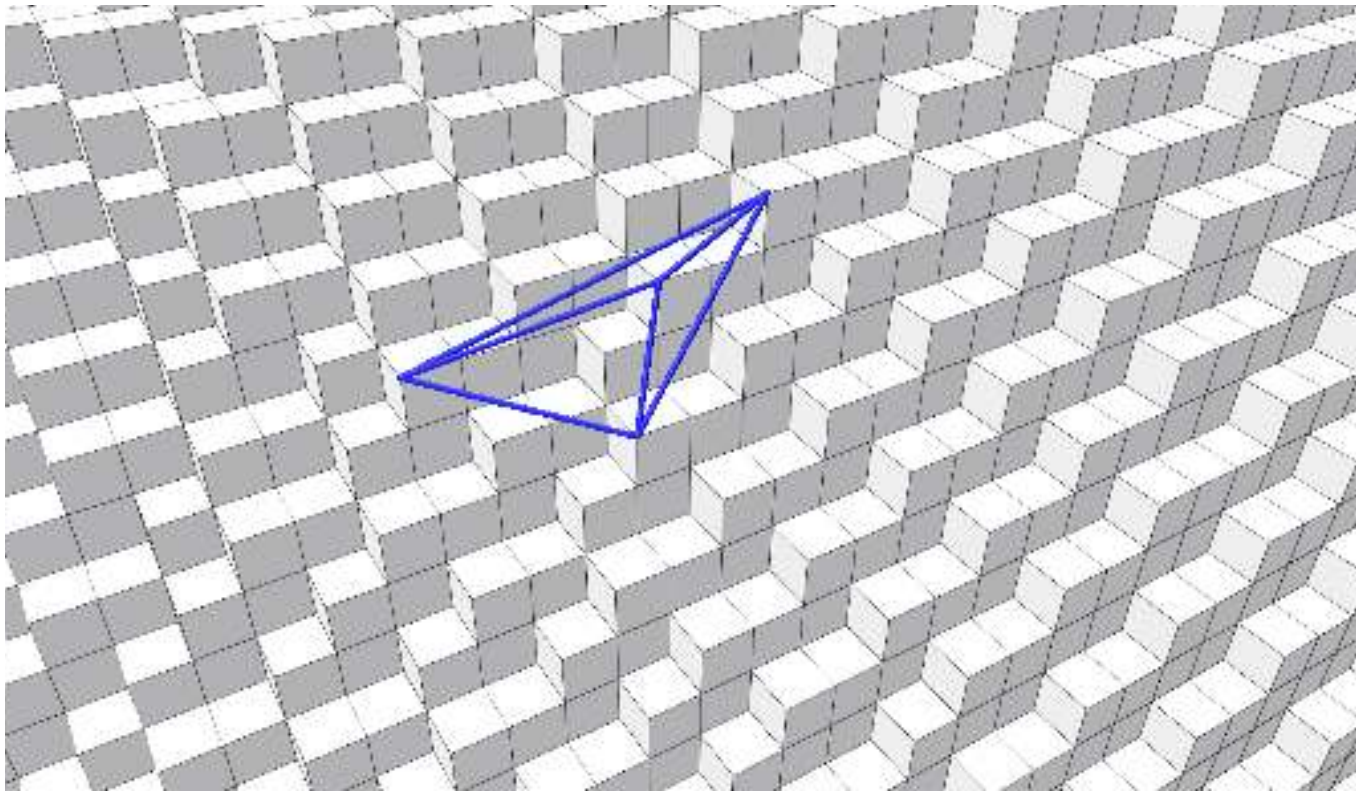

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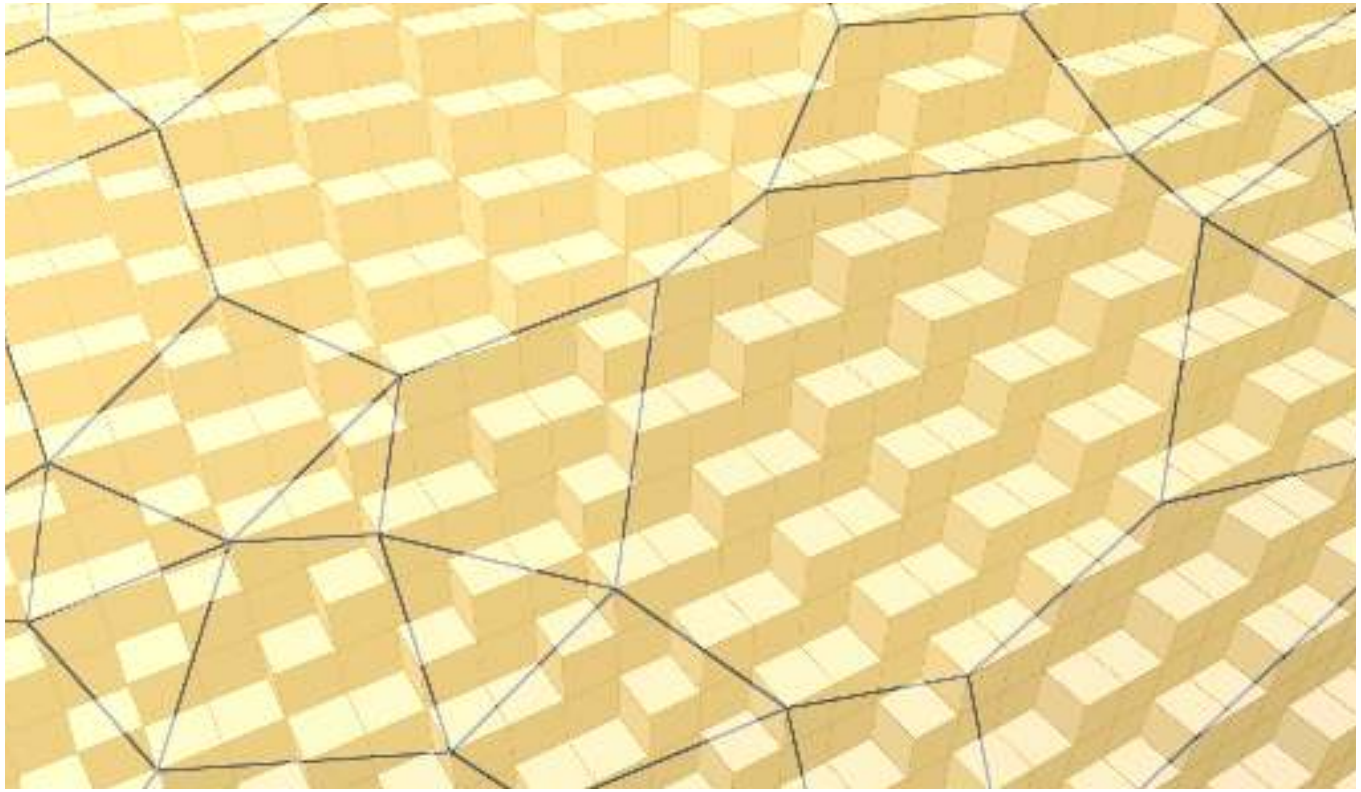

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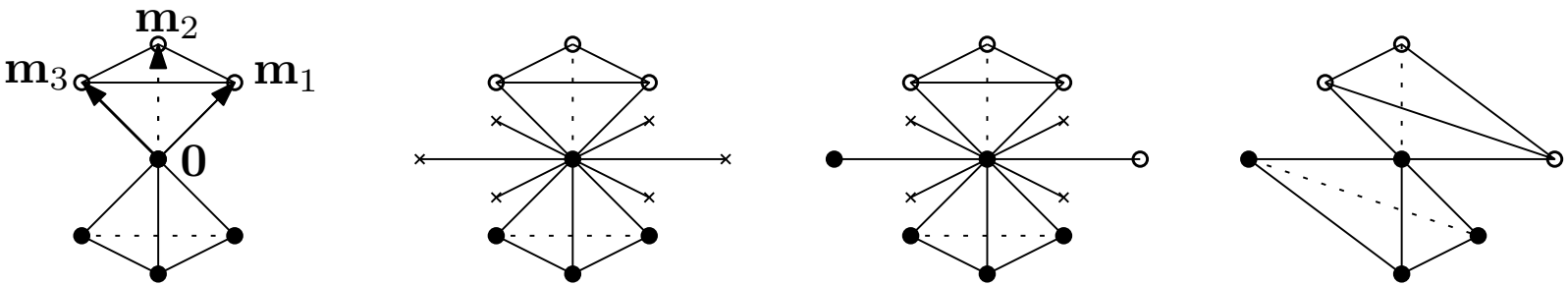
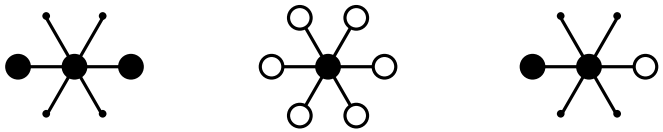
# Algorithm IsExtremal

- Invariant: $\det(\mathbf{M}) = 1$, $-\mathbf{m}_k \in Z_K, \mathbf{m}_k \notin Z_K$
- Loop these steps:




vectors $\mathbf{m}_k$ go up, $\mathbf{m}_k - \mathbf{m}_{k\pm1}$ in $Z_K$? if yes, $\mathbf{m}_k \leftarrow \mathbf{m}_k - \mathbf{m}_{k\pm1}$

- possible configurations at the six points $\mathbf{m}_k - \mathbf{m}_{k\pm1}$:



No Yes Probing

# Algorithm termination

**Theorem**

*If $Z_k$ is finite, algorithm* IsExtremal *terminates after a finite number of iterations.*

**Theorem**

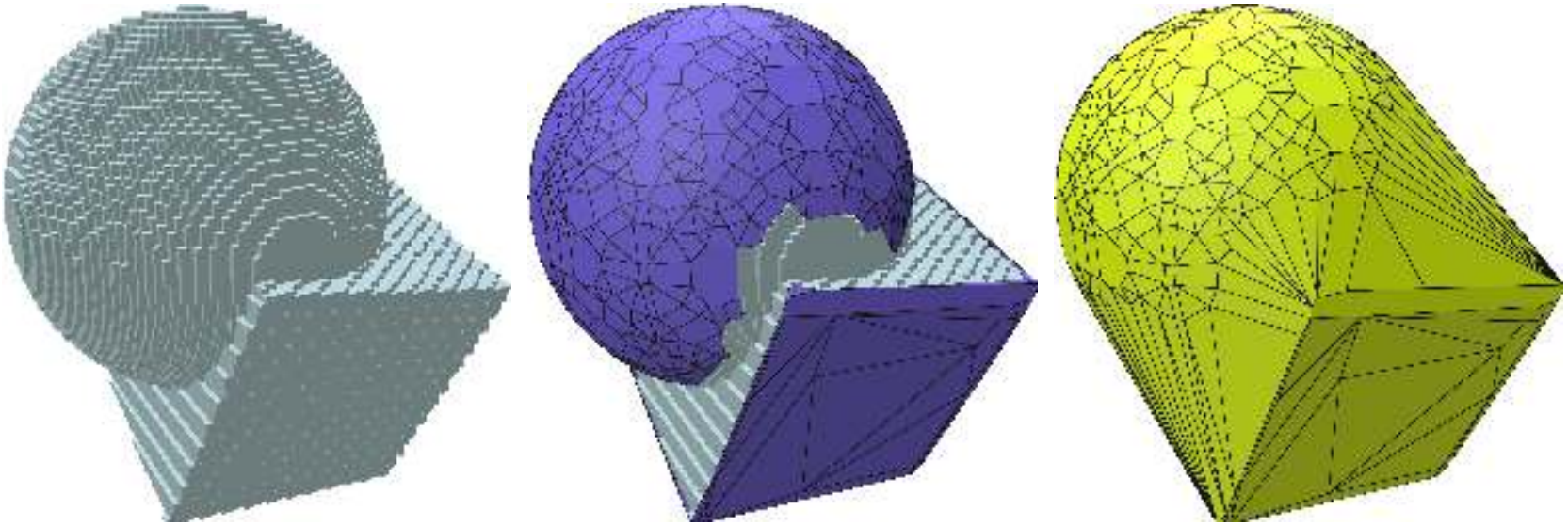
*If the point to test* $\mathbf{x}$ *is a vertex of* $\mathrm{Cvxh}\,(Z_k)$*, algorithm* IsExtremal *returns* Yes *after at most* $n$ *iterations, with* $n \leqslant 2\sqrt{3}A$ *and* $A$ *the total area of the facets of* $\mathrm{Cvxh}\,(Z_K)$ *incident to* $\mathbf{x}$*.*

# How good is the probing algorithm as a filter?

- efficient on digitizations of smooth shapes (here ellipsoid) with gridstep $h$
- $n_{init}$: number of salient corners
- $n_{final}$: corners labeled as extremal by algorithm IsExtremal
- $n_{\mathrm{Cvxh}(Z_K)}$: expected number of vertices of $\mathrm{Cvxh}\,(Z_K)$

| grid step | $\#Z_K$ | $n_{init}$ | $n_{final}$ | $n_{\mathrm{Cvxh}(Z_K)}$ |
|---|---|---|---|---|
| 0.5 | 984 | 112 | 112 | 112 |
| 0.1 | 24.808 | 2.032 | 1.128 | 1.128 |
| 0.05 | 99.448 | 7.784 | 3.064 | 3.064 |
| 0.01 | 2.488.104 | 186.664 | 33.864 | 33.784 |

# A few results (convex zones)



| shape | $\#Z_K$ | $n_{init}$ | $n_{final}$ | #facets | time(ms) |
|---|---|---|---|---|---|
| cps | 34036 | 3681 | 991 | 959 | 2529 |
| torus-knot-128 | 96622 | 15196 | 2924 | 2752 | 29321 |
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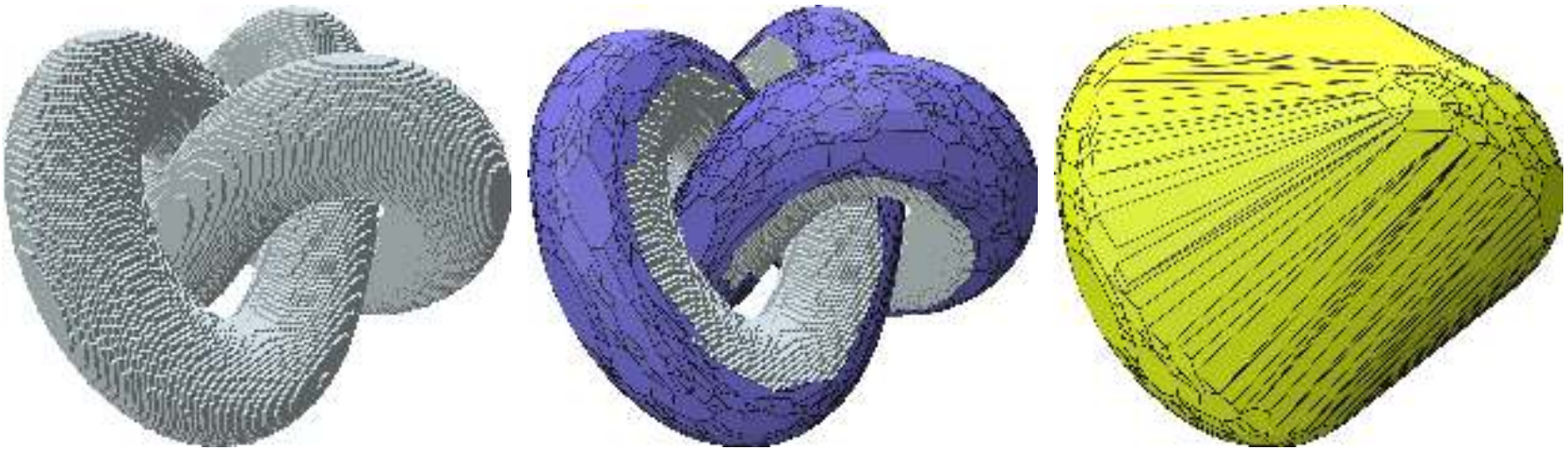

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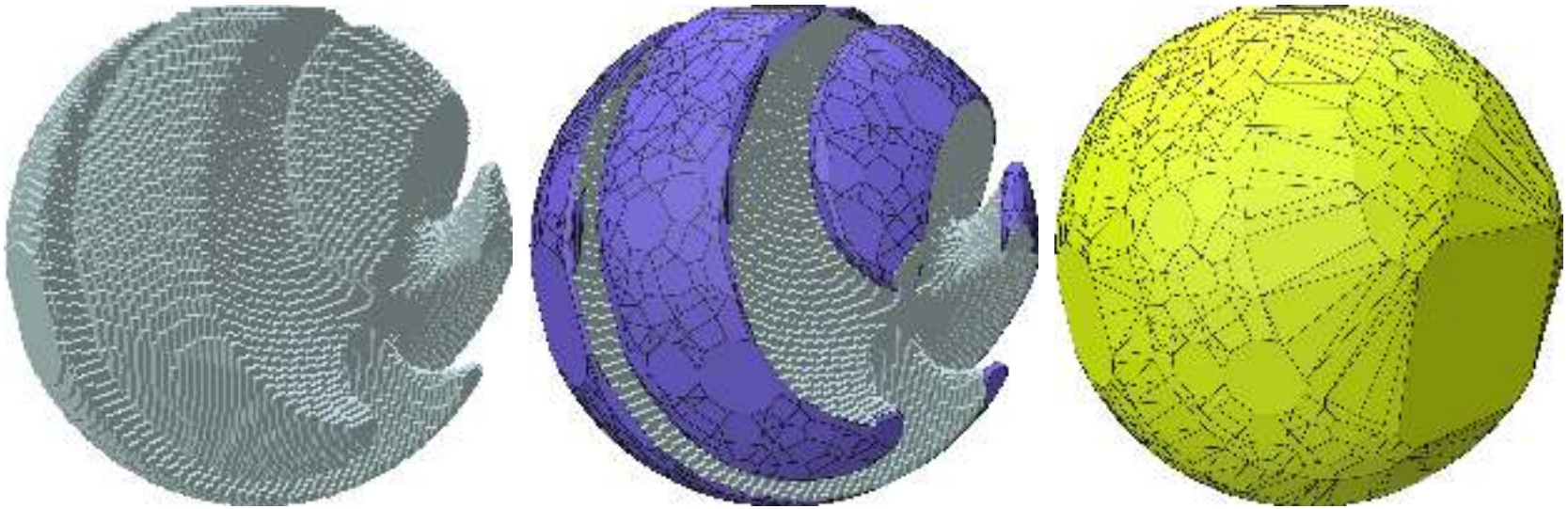

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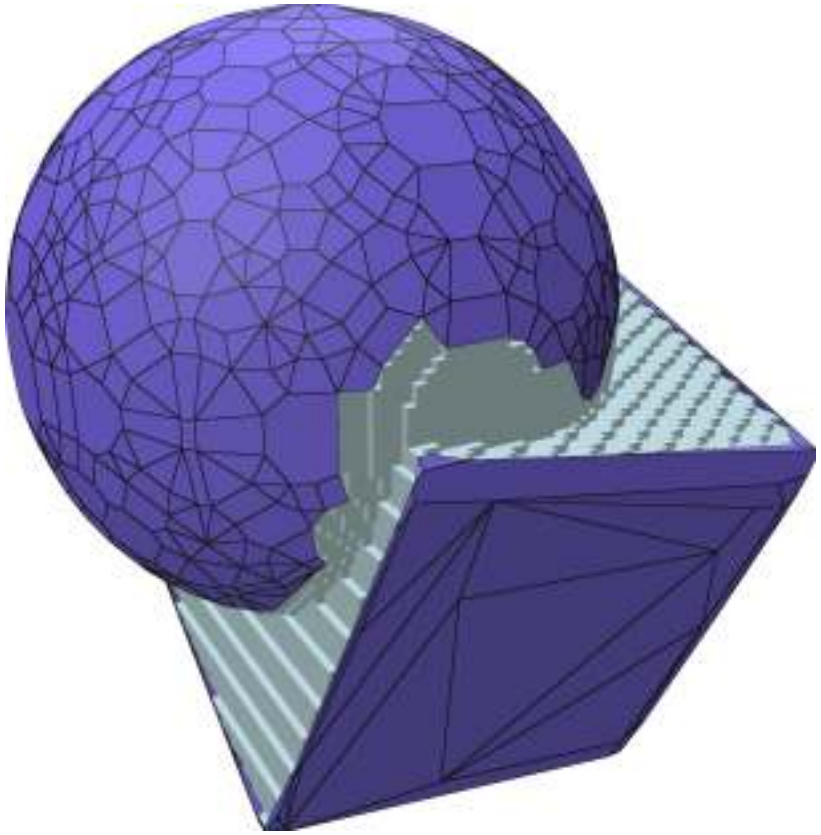
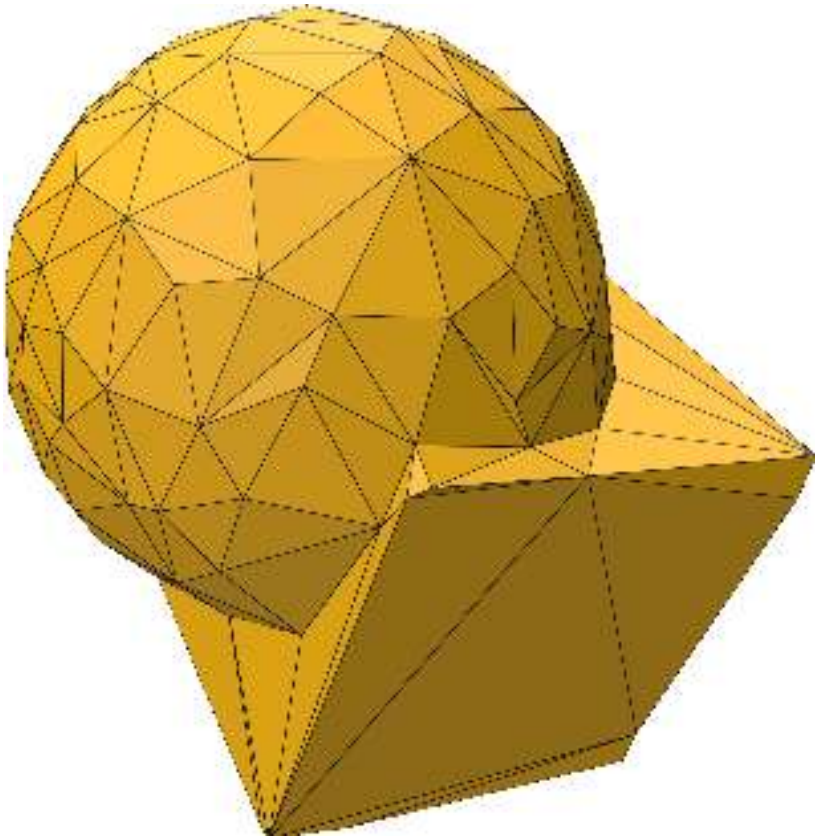
# Comparison with mesh simplification



our method



mesh simplification

Both methods are at Hausdorff distance 1 from digital surface.

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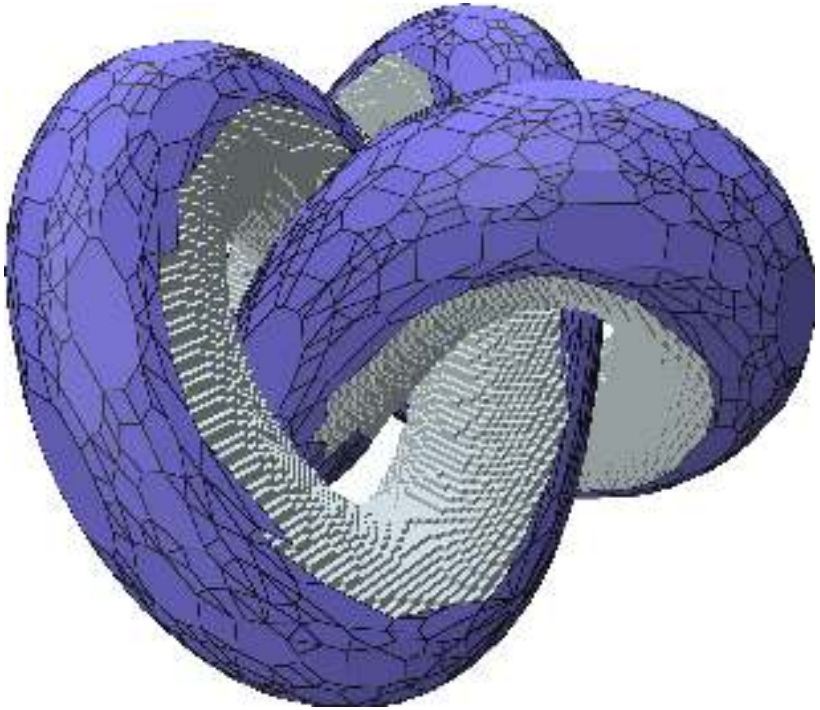

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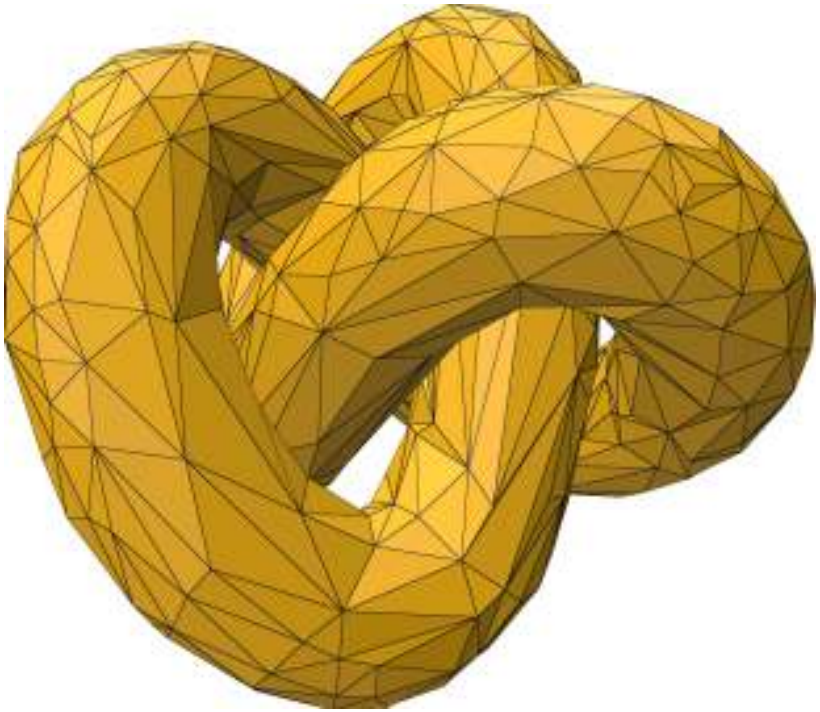

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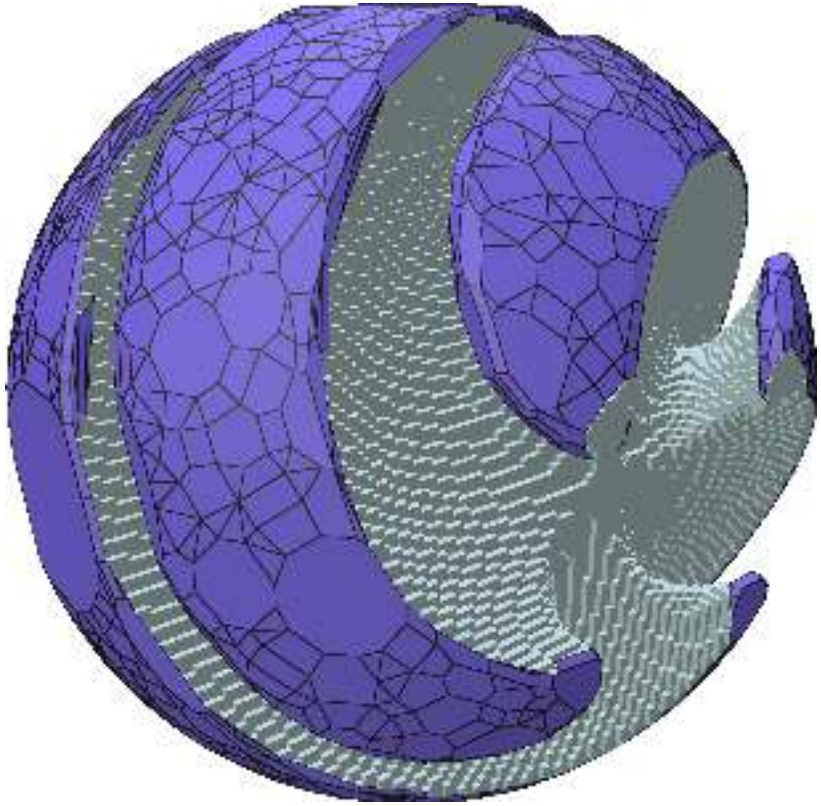

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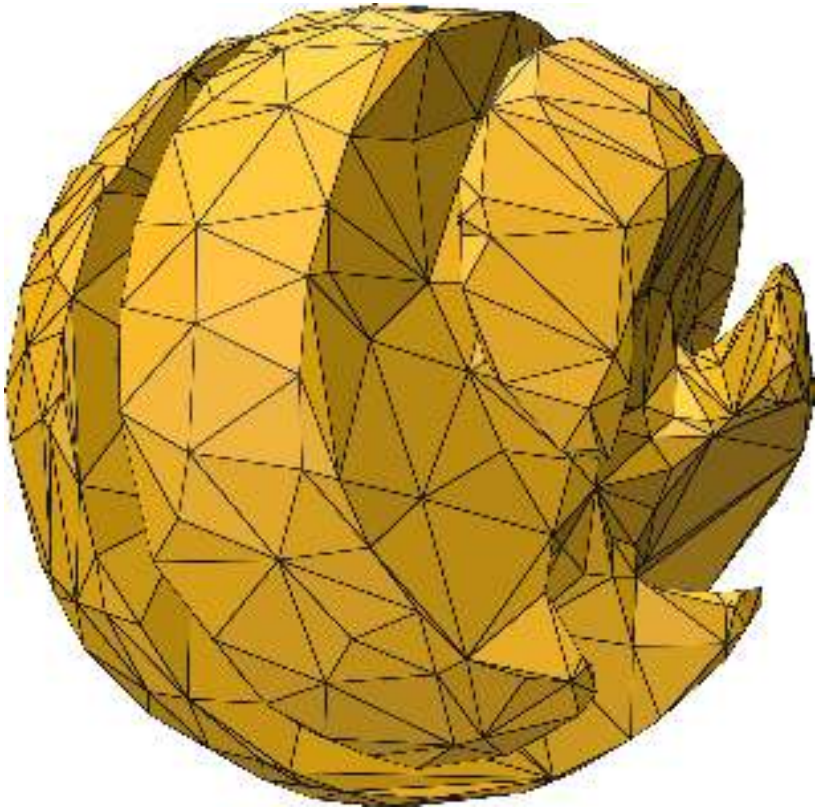

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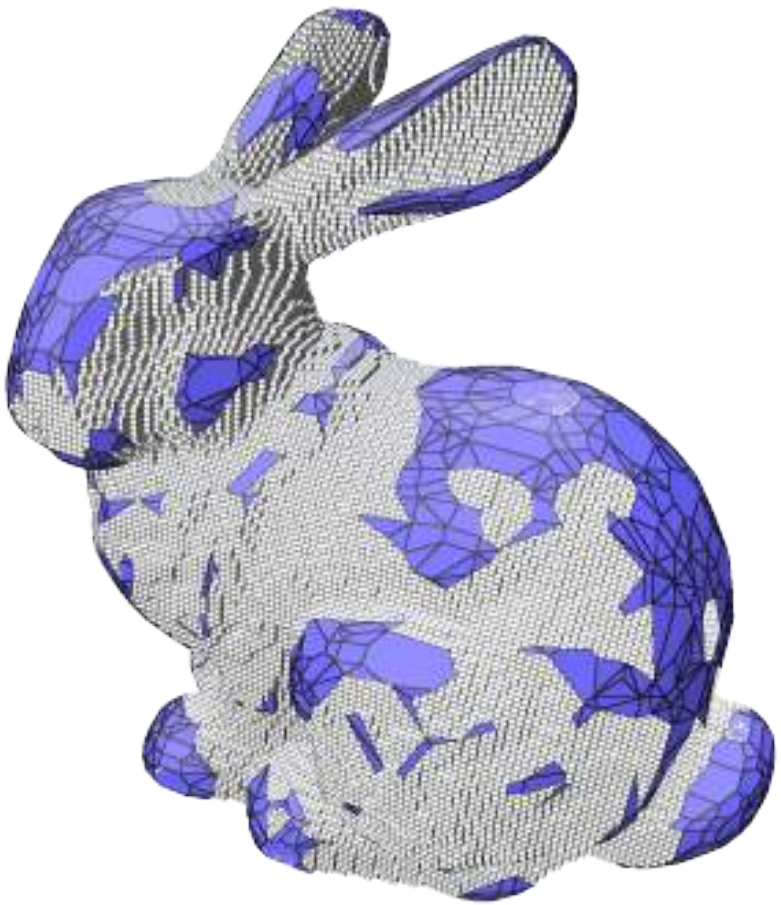

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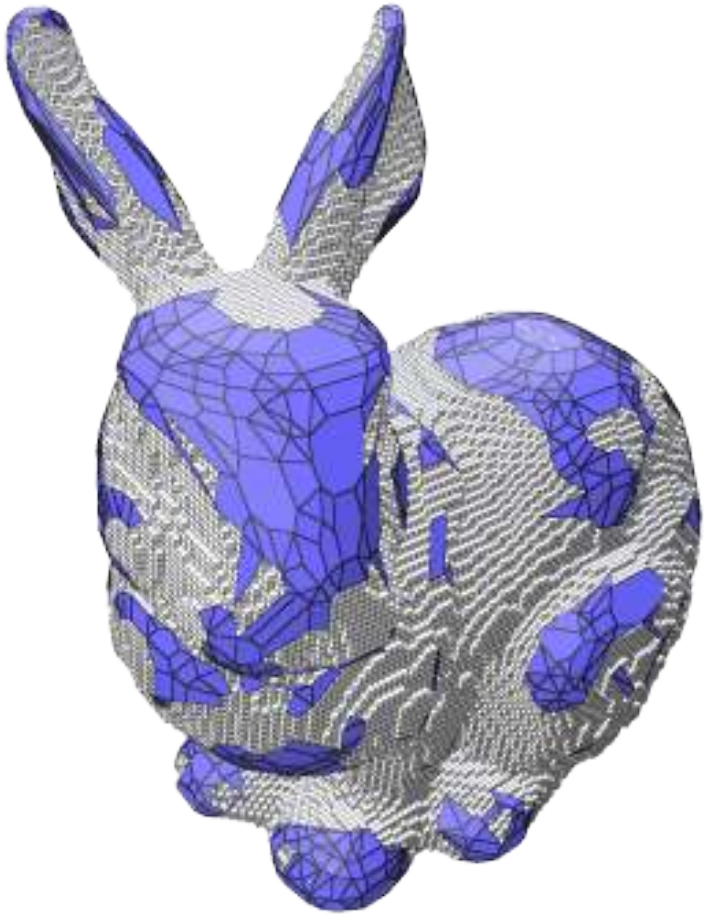

# Conclusion and perspectives

**Contribution**

- local definition of convexity/concavity through convex hulls of visibility cones
- algorithm that reconstructs the locally convex/concave vertices, edges and faces, ...
- fast probing algorithm to identify at 99% extremal points

**Perspectives**

- speed up: compute visibility in a coarse-to-fine way
- How to triangulate neither convex nor concave parts?
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Thank you for your attention !